The Effect of Inequality on Redistribution: An Econometric Analysis *

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Abstract

Using a quarter century of U.S. state and federal tax and transfer data, we estimate how shifts in the market income distribution pass through to the distribution of post-tax, post-transfer disposable income. A system of equations yields the marginal effect of moving market income share from any quintile to any other on the *entire* disposable distribution. We address reverse causality by instrumenting with state-level exposure to international trade, commodity-price, and national industry-demand shocks. The system's attenuation of market income shifts is hump-shaped in quintile rank, peaking sharply at the middle quintile, consistent with the median voter theorem and what Stigler labeled as Director's Law. This works largely through large, systematic spillovers onto quintiles that neither gain nor lose, also favoring the middle, which we term the greedy median voter. The shape of this pass-through shifts with economic conditions. Where market inequality has run higher, the middle-favoring peak flattens and the bottom quintile is left more exposed. As average real disposable income per capita rises the protection of the bottom likewise recedes, even as the middle peak holds firm. We complement these results with an insurance-coalition analysis.

Keywords: Inequality, redistribution, voting, political economy, taxes, transfers

JEL codes: D63, D72, H23

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1 Introduction

The last several decades have witnessed a sharp increase in income inequality throughout the developed world.[1] In recent years, this trend has become a major issue of public interest, and the surrounding debate has underscored the influence of distributional concerns on the design of tax codes, transfer programs, and fiscal policies. The nature of this influence is the topic of this paper. We empirically investigate the effect of changes in income inequality on redistributive tax and transfer policy, assessing the role and importance of various political-economic theories in the patterns we observe.

Using Census data from the ASEC supplement to the Current Population Survey, we build and estimate a regression model which maps the distribution of market income to the distribution of post-tax, post-transfer disposable income in U.S. states over the last quarter century. Each state-year population is first ranked by market income and broken up into five quintiles of equal size. A system of equations then associates quintiles' market income shares with a corresponding set of final, disposable income shares (the functional form ensures that the predicted shares always sum to one). To isolate pure changes in market income inequality, we control for the level of income, the business cycle, and demographics in the regression. We also account for reverse causality, using proxies for exposure to international commodity price shocks, international trade shocks, and national industry demand shocks as instruments.[2]

Once the model is estimated, it is possible to compute the marginal effect of a shift

[1]See Stand and Rising (2011). Autor, Katz, and Kearney (2008) highlight the large difference in trends within the distribution, as the 90/50 ratio accounts for most of the rise in inequality, whereas the 50/10 ratio has been fairly stable. There is a growing body of literature addressed to the wage premium gap and related issues. The seminal Goldin and Katz (2009) history frames the evolution as a "race between education and skill-biased technical change" and thus the increased demand for highly skilled relative to lower skill supply changes from education. Their analysis is updated by Autor, Goldin, and Katz (2020) to account for convex returns to schooling. Boskin and Lau (2000) estimate that technical change for the G-7 countries has been both capital and human capital biased, what they label generalized Solow neutral technical change.

[2]This set of instruments is further supplemented by forecasts of the endogenous variables based on lagged values or national trends, and various extensions consider nonlinear specifications, interaction terms, etc.

of market income share from one quintile to another on the entire disposable income distribution. Because such shifts are not all equally likely, we examine the "experiments" which are most representative of recent national and state histories in detail. We then broaden the scope of the analysis by summarizing and distilling the full set of estimates, the marginal effects for each quintile pair and every possible loss-gain scenario.

The first question is whether the so-called "redistribution hypothesis" holds. As inequality increases, does redistribution increase to compensate?[3] Confirming early studies, we find that, indeed, taxes and transfers attenuate shifts in the market income distribution. Quintiles which lose market income share experience a smaller drop in their disposable income share, and quintiles which gain market income share experience less growth in their disposable income share. This is unsurprising, but in contrast to the existing literature, the richness of the model allows us to probe further. For the first time, we investigate not only whether taxes and transfers stabilize disposable income inequality, but how such stabilization occurs and to what degree the responses depend on the way in which the market income distribution is transformed, the quintiles involved, etc. In particular, we show that the degree of attenuation (as described above) is initially increasing in a quintile's percentile rank, peaks at the middle quintile, and then falls for higher income quintiles. In other words, tax and transfer policy is most sensitive to the fortunes of the middle class, consistent with the median voter theorem.[4] We also find that redistribution has significant spillover effects on quintiles which neither gain nor lose market income in the experiments. That is, rather than simply neutralize or reverse shifts in the market income distribution, government intervention also spreads and dis-

[3]This is also usually the result of the benevolent social planner of optimal tax Mirrlees models, although these explore the relationship of inequality in the ability to earn income, not the ex- post realization of market income which also reflects effort. Heathcote, Storesletten, and Violante (2020) find the utilitarian social planner chooses less progressivity in response to widening skill price dispersion reflecting technical change. This offsets more progressivity in response to larger residual inequality, leaving overall optimal progressivity unchanged for the U.S. data they calibrate for 1980-2016.

[4]For example, Meltzer and Richard (1981) apply the classic median voter theorem of Hotelling (1929) and Black (1948) to a model economy with income inequality, redistributive taxation, and endogenous labor supply.

perses gains and losses across the *entire* income spectrum. Under a fixed net tax schedule (where "net taxes" are taxes net of transfers, and "fixed" means invariant to distribution shocks), such spillovers would be small, so we interpret this as some combination of active policy responses and more complex policy formulas in practice. Finally, the results indicate that these patterns, as well as, to a large extent, the estimates themselves, are largely invariant to most of the interaction terms we include in the regressions. Two of them are exceptions. The level of real disposable income per capita has considerable effects, operating mainly through the cross-quintile spillovers. And the history of market income inequality matters; higher past inequality flattens the middle-quintile peak and shifts protection toward the second quintile (see Section 5.2). These patterns are robust: the middle-quintile peak and the exposed bottom survive variation in the instrument set, the sample of states, the time period, and the price deflator (Section 6).

To our knowledge, no other paper recovers the entire mapping from the market income distribution to the disposable income distribution quintile by quintile, respecting the adding-up constraints (shares sum to 1 and share changes sum to 0). The closest empirical tests of Director's Law instead relate a single index of inequality to a summary measure of redistribution, rather than tracing one full distribution into the other. This framework opens up various new avenues of inquiry, not all of which are covered in this paper. For instance, the framework lets us examine which quintiles' disposable-income shares move together, an insurance-coalition structure we take up below.

Moreover, we are aware of only one other paper which instruments for structural changes in the market income distribution (Pamp and Mohl, 2009). Because taxes and transfers have obvious implications for work and other incentives and therefore influence the market income distribution, the lack of any correction for reverse causality is an important limitation of previous work. The regional nature of our data makes this issue surmountable. Drawing on methods and insights in Bartik (1991), Autor, Dorn, and Hanson (2013), Voorheis, McCarty, and Shor (2015), Asquith et al. (2019), and Kehrig and Ziebarth (2017), we exploit local heterogeneity in industry composition and state-level exposure to national and international trends to identify exogenous shocks to the market income distribution. Instrumenting proves consequential, not cosmetic. The correction for reverse causality is precisely what turns the conventional picture, broad protection for the bottom of the distribution, into the median-voter pattern, in which the middle quintile is the most insulated. Uninstrumented (seemingly unrelated regression) estimates recover the former. Our instrumented estimates recover the latter, and we use the contrast throughout as a window on the underlying political economy.

Related Literature. Although a number of important papers have tested the proposition that greater inequality leads to more redistribution, until recently, researchers have lacked the data to construct comprehensive income measures, let alone characterize the distributions of both market and disposable income. As a result, many studies are based on limited proxies for inequality, the level of redistribution, or both. For instance, Meltzer and Richard (1983) uses earnings data, failing to account for capital income and the sorting of wage earners into households, while Alesina and Rodrik (1994) substitutes inequality in land ownership for income inequality. Several papers, such as Perotti (1996) and Bassett, Burkett, and Putterman (1999), measure redistribution as the ratio of transfer spending to GDP, a statistic which does not capture the distribution of transfers or the effect of progressive taxation. These papers also relate redistribution to disposable income inequality, finding, if anything, a negative relationship, an unsurprising result given that the direct effect of the former is to reduce the latter.

More recent studies, beginning with Milanovic (2000) and including Kenworthy and Pontusson (2005), Mahler (2008), Milanovic (2010), and Ostry, Berg, and Tsangarides (2014), address many of these issues. Utilizing national survey microdata, harmonized across countries and over time, these papers disentangle the market and disposable income distributions, measuring redistribution either as the difference in Gini coefficients or the income share gained by a lower quantile. They find that taxes and transfers do, in fact, play a stabilizing role (i.e. greater inequality is associated with more redistribution). Several recent studies test this Meltzer-Richard relationship more directly. Gründler and Köllner (2017) estimate a system-GMM panel of redistribution on market inequality across countries and, like us, find the relationship driven by the middle of the distribution, the third-quintile income share raises redistribution while top shares resist it, though theirs is a cross-country correlation rather than a within-country causal estimate. Borge and Rattsø (2004) provide a sub-national counterpart, showing across Norwegian municipalities that a more equal income distribution shifts the local tax mix away from redistributive policy instruments, consistent with Meltzer-Richard but without addressing the endogeneity of the income distribution that we instrument.

Recent quasi-experimental work confirms that distributional policy is causally consequential: Hope and Limberg (2022) find that major tax cuts for the rich raise the top one percent's pre-tax income share by roughly a percentage point over five years, with no measurable gains to growth or employment. Closest to our own exercise is Pamp and Mohl (2009), to our knowledge the only prior comprehensive empirical test of Director's Law. Using Luxembourg Income Study survey data on income-decile share gains across 22 (largely European) OECD countries over 1971–2005, they find that rising inequality redistributes toward the middle class at the expense of both the top and the bottom. We reach the same middle-favoring conclusion, but on quite different ground. Within a single country, across U.S. states, rather than across heterogeneous national systems; on CPS income with the detailed in-kind valuation of taxes and transfers, and the value of the imputed income of homeownership, described in Section 3, rather than harmonized cross-national survey aggregates; with external trade, commodity-price, and industry-demand shocks as instruments, rather than internal lags alone; and, above all, recovering the full quintile-by-quintile mapping, the shape of attenuation and the cross-quintile spillovers, rather than relating a single scalar index of inequality to a summary measure of redistribution.

The renewed interest in distributional issues has generated important investments in data development. The World Inequality Database, for example, is increasingly used to describe trends and analyze issues, e.g. Blanchet, Chancel, and Gethin (2022). Often they combine survey and administrative data, even augmented to try to be consistent with national income accounts data. Of course, each step requires assumptions. While some harmonization is accomplished, it is not complete, given differences in the survey methods and data collection choices of different statistical agencies. To create annual time series data, considerable imputations and extrapolations are sometimes necessary. The more countries included and the further back in time, the more pronounced these issues become. What's more, in-kind income, a category which includes the imputed rent from owner-occupied housing and publicly financed healthcare (among other income sources received in the form of goods and services rather than cash), either is excluded to maintain consistency or requires extensive interpolation and extrapolation.

In contrast, our state-level data draws upon surveys or administrative data sets from the same statistical agencies, consists of over 1000 observations, includes the same set of years for each state (in the main analysis without gaps), and accounts for in-kind sources of income. Of course, the data, and our income definition, are not perfect. For example, the underreporting or omission of capital gains is especially important at the very top. Armour, Burkhauser, and Larrimore (2014) showed that the use of realized capital gains (based on tax unit data) overstate the increase in inequality. A case can also be made that consumption is a better measure of living standards and permanent income for many households (Attanasio, Battistin, and Padula, 2011) and/or that longer term measures should be used, or at least added, to the discussion (e.g. the classic book by Vickrey (1947) and the important body of work by Auerbach and Kotlikoff (1987, 1998) and Auerbach, Gokhale, and Kotlikoff (1994)). However, consumption data are often less comprehensive than income data in household surveys.[5] In any case, the trends from

[5]A laudable exception is the more recent waves of the Panel Study of Income Dynamics, but these data are too limited for our analysis.

the 1980s to the Great Recession appear to be fairly similar.[6]

Analyses of the distribution of economic (or more general) well-being must make many choices, including: 1) the unit of account, e.g. households, families, individuals, tax units; 2) the time period of measurement, e.g. annual, multi-year, lifetime, and in any trend analysis; 3) the measure of well-being, e.g. market income, disposable income, consumption, wealth, and how comprehensively to measure it, for analytical, policy, statistical or data availability reasons; 4) the ranking (and possibly re-ranking) of households methodology used; 5) closely related to 1) and 3), adjustments for unit size, cost of living and any trends therein.[7] Of course, any (one or more) of these may significantly affect the measured outcome.[8] The literature contains several excellent analyses of the pros and cons of, and inequality trends in, alternative measures, e.g. Deaton and Muellbauer (1986), Fisher, Johnson, and Smeeding (2013), Meyer and Sullivan (2023), Blank and Greenberg (2008), Slesnick (1993), Hutto, Waldfogel, Kaushal, and Garfinkel (2011), and Rose (2020). We make choices that seem sensible to us, for the purposes of our analysis, given the constraints involved, surprisingly including huge gaps in recent years in available data on Medicaid discussed below.

Also relevant are the recent important attempts to assess the usefulness of the widely used median voter model. For example, Acemoglu, Naidu, Restrepo, and Robinson (2015) examine a series of factors that lead to different or ambiguous outcomes than the median voter model. Mulligan, Gil, and Sala-i Martin (2004) emphasize the degree of competition rather than the voting processes. Alesina and Giuliano (2011) explore preferences for redistribution, using social and value surveys of individuals and empha-

[6] However, it appears they diverged (consumption inequality decreasing, income inequality continuing to rise) after 2007. See Meyer and Sullivan (2013). Also see Krueger and Perri (2006) for an interesting attempt to analytically relate the two.

[7] Rose (2018) and Early (2018) discuss many of these measurement issues and their implications.

[8] For example, see the debate over how much top income shares have truly risen: Piketty, Saez, and Zucman (2018) and Saez and Zucman (2020) find a near-doubling of the top one percent's pre-tax share since 1980 with taxes and transfers offsetting only a small fraction of the rise, whereas Auten and Splinter (2024) (see also Auten and Splinter, 2021, Splinter, 2020), correcting for tax-base changes, missing income, and the tax-unit-versus-household distinction, find a far smaller increase and essentially flat after-tax top shares, methods whose implied top-share growth differs by a factor of almost five.

size personal histories and cultural factors. Marechal, Cohn, Yusof, and Fisman (2023) compare realized redistribution outcomes in a cross section of countries and conclude the preferences of the lower social economic groups are most decisive. Our results with a quite different methodology, using US data and instrumenting to account for reverse causality of tax and transfer effects on market income and therefore putatively causal, are more consistent with the median voter model. The difference may be explained in the results of Benabou and Tirole (2006), whose model produces two equilibria, labeled "American" and "European." The former is characterized by the belief that effort pays off and has more laissez faire outcomes, whereas the latter is characterized by pessimism that effort pays off and results in a welfare state. That Pamp and Mohl (2009) document the same middle-favoring redistribution in a largely European sample, while we find it within the United States, suggests the Director's-Law pattern operates across both equilibria, even where their overall levels of redistribution differ.

There are many important extensions to understanding both the market distribution of income and its relationship to income levels, on the one hand, and preferences for redistribution (Alesina and Giuliano, 2011), even how voters think about redistribution (Stantcheva, 2021), and political outcomes, on the other. While our analysis here focuses on the usual current market and disposable incomes, income mobility, income instability (Ganong, Noel, Patterson, Vavra, and Weinberg, 2024), intergenerational mobility (Alesina, Stantcheva, and Teso, 2018), expectations of future income, taxes and transfers, the effects of marginal incentives as opposed to current average taxes and transfers (Barro and Redlick, 2011), longer-term, even lifetime, measures of income and net taxes, and rankings based thereon (Auerbach, Kotlikoff, and Koehler, 2023), and other important dimensions could eventually be valuable complementary supplements/extensions to our analysis.

The paper is organized as follows. Section 2 presents the model and introduces notational definitions of the key variables. Section 3 describes the choice of sample and the income definition in greater detail and specifies how each variable is constructed from the relevant data sources. Sections 4 and 5 present the results, Section 6 reports robustness checks, and Section 7 assesses model fit and instrument validity. Section 8 discusses directions for further research, and Section 9 concludes. A full set of sensitivity analyses is presented in the Appendix.

2 The Model

2.1 The Mapping from Market Income to Disposable Income

Let the term "quintile j" refer to the j^{th} quintile, counting from the bottom, of the corresponding (either disposable or market) income distribution. For example, "quintile 4" refers to the 61$^{\text{st}}$ through the 80$^{\text{th}}$ percentiles. Write disp^j_{it} and mkt^j_{it} for the mean disposable income (market income after taxes and transfers) and the mean market income, in dollars, of quintile j in state i and year t. The corresponding income shares are the disposable share $D^j_{it} = \text{disp}^j_{it} / \sum_m \text{disp}^m_{it}$ and the market share $m(j)_{it}$, defined analogously. We model each quintile's disposable income as a linear function of the market income distribution, a vector of controls, and state and year effects:

$$\text{disp}^l_{it} = \alpha^l_i + \gamma^l_t + \varphi^l + \beta^l \cdot \boldsymbol{M}_{it} + \theta^l \cdot \boldsymbol{Y}_{it} + \epsilon^l_{it}, \qquad l = 1, 2, \ldots, 5, \tag{1}$$

where $\boldsymbol{M}_{it}$ is the vector of quintile mean market incomes, $\boldsymbol{Y}_{it}$ a vector of controls, α^l_i, γ^l_t, and φ^l are state effects, year effects, and an intercept, and ϵ^l_{it} is equation l's error. There is one equation for each of the five disposable quintiles.[9]

The specification also includes aggregate market income per capita, which fixes the overall level of income so that the quintile market-income terms isolate purely distributional shifts. Because aggregate income per capita is the average of the five quintile

[9] Because of administrative costs in the tax and transfer system, government purchases, and potentially large compliance and incentive-distortion costs, total disposable income is less than total market income.

means, one quintile must be dropped from $\boldsymbol{M}_{it}$ as a "base" to avoid collinearity. With quintile l excluded, an increase in mkt^n holding aggregate income fixed represents a transfer of market income from the base quintile l to quintile n. The choice of base is immaterial, the estimates under one base determine those under any other, and the effect of a transfer from quintile i to quintile j is equal and opposite to that of the reverse transfer (Appendix A).

In summary, there are 20 possible "experiments" in which one of the five quintiles loses market income share to one of the other four, and each potentially affects the disposable income shares of all five quintiles, for a total of one hundred marginal effects. Because the effect of a shift from quintile i to quintile j is equal and opposite to that of the reverse shift, fifty of these are independent.

2.2 Shares as the Object of Interest

The ultimate object of interest is distributional: the effect of a shift in the market income distribution on the disposable income distribution, each summarized by quintile shares. Two observations let us read this share object directly off the dollar system in (1).

First, income shares, not dollar amounts, are what we care about, and shares are invariant to the price level. Within a given state and year, every household's income is measured in the same dollars, so any common deflation, whether temporal or spatial (regional price parities), divides each quintile's income by the same state-year scalar. That scalar cancels in every share and leaves the ranking of households unchanged, so the within-state-year distribution we model is numerically identical in nominal and in real terms. The price level can therefore enter our analysis only through the real-income-per-capita interaction and through comparisons across state-years, never through the share variables themselves. We accordingly estimate (1) in nominal dollars, which keeps the underlying magnitudes transparent, and we confirm in Section 6 that re-estimating the system in fully deflated terms, applying both the PCE temporal deflator and BEA regional price parities, leaves the recovered attenuation structure, the middle-quintile peak, and the cross-quintile spillovers essentially unchanged. The most exacting deflation would go further still, letting the consumption basket—and thus the price index—vary across income groups, since inflation is not distributionally neutral (Carloni, 2025); we regard incorporating such group-specific indices as a natural refinement for future work.

Second, the share object is recovered from the dollar coefficients by an exact identity. The disposable share is $D^l = \mathrm{disp}_l / D$, where $D = \sum_m \mathrm{disp}_m$ and $M = \sum_m \mathrm{mkt}_m$ are aggregate disposable and market income per capita. The quotient rule gives the exact response of this share to a market shift,

$$\Delta D^l = \frac{\Delta \mathrm{disp}_l - D^l \, \Delta D}{D}, \qquad \Delta D = \sum_m \Delta \mathrm{disp}_m,$$

an identity, not an approximation. Since (1) is linear, the dollar response $\Delta \mathrm{disp}_l$ equals the estimated coefficient β_l and is constant. Evaluating this share map at every state-year it and averaging gives the average marginal effect (AME)

$$\begin{aligned} \widehat{AME}_l &= \frac{1}{N} \sum_{it} \frac{M_{it}}{D_{it}} \Big(\beta_l - D^l_{it} \sum_m \beta_m \Big) = w_0 \beta_l - \bar{w}_l \sum_m \beta_m, \\ w_0 &= \frac{1}{N} \sum_{it} \frac{M_{it}}{D_{it}}, \qquad \bar{w}_l = \frac{1}{N} \sum_{it} \frac{M_{it}}{D_{it}} D^l_{it}. \end{aligned} \tag{2}$$

Here β_l is the dollar GMM coefficient—the response of quintile l's disposable income to the transfer—$\sum_m \beta_m$ is the induced change in aggregate disposable income, D^l_{it} is quintile l's disposable share, and the factor M_{it}/D_{it} rescales the per-dollar response into the response of a *share* to a one–percentage-point shift of market-income share; w_0 and $\bar{w}_l$ are the corresponding sample averages, which enter as fixed data weights. Each recovered effect is thus one experiment—one quintile cedes a percentage point of market-income share to another—and $\widehat{AME}_l$ is the average response of quintile l's disposable share. This map is linear in the GMM coefficients (income levels enter only as fixed data weights), so its standard errors are exact by the delta method, computed from the same cluster-robust covariance as the dollar estimates, and the recovered effects satisfy the adding-up constraint $\sum_l \widehat{AME}_l = 0$ by construction. All share marginal effects, attenuation structures, and spillovers reported below are these recovered objects, with delta-method standard errors throughout.

We estimate the system (1) in levels, where the market-income quintile means are strongly instrumented by their own deep lags, and recover the share objects analytically. Because levels and shares are exact transforms, nothing is lost, and the first stage is strong (Section 7).

The recovered effects in (2) do more than summarize the unweighted attenuation structure: because both income distributions are described by *shares*, the same effects map *any* contemplated change in the market distribution into its disposable counterpart. Collect them into a single 5×5 matrix $\boldsymbol{B}$, whose (l, n) entry $\widehat{AME}_l(n)$ is the recovered share effect in (2) on quintile l's disposable share of a one–percentage–point transfer of market–income share into quintile n. Three properties organize how this matrix is used. First, because market and disposable shares each sum to one, every feasible market reallocation is zero–sum and the disposable changes it induces sum to zero, so each column of $\boldsymbol{B}$ inherits the adding–up constraint of (2) (equivalently, the disposable–share responses in each row of Table A3 sum to zero),

$$\sum_{n=1}^{5} \Delta m^n = 0, \qquad \sum_{l=1}^{5} \widehat{AME}_l(n) = 0, \qquad \text{so} \qquad \sum_{l=1}^{5} \Delta D^l = 0. \tag{3}$$

Second, since the dollar system (1) is linear, a reallocation of the market–share vector $\Delta \boldsymbol{m} = (\Delta m^1, \ldots, \Delta m^5)'$ maps into the disposable–share response through these estimated effects,

$$\Delta \boldsymbol{D} = \boldsymbol{B}\, \Delta \boldsymbol{m}, \qquad \mathbf{1}'\boldsymbol{B} = \mathbf{0}'. \tag{4}$$

Third, any such $\Delta \boldsymbol{m}$ is a non–negative combination of the elementary one–point transfers $\boldsymbol{e}_{i\to j} = \boldsymbol{e}_j - \boldsymbol{e}_i$ (moving one point of share from Q_i to Q_j), so the per–unit disposable response under any scenario is a normalized weighting of the estimated experiments,

$$\Delta \boldsymbol{D}(\boldsymbol{\omega}) = \boldsymbol{B} \sum_{i\to j} \omega_{i\to j}\, \boldsymbol{e}_{i\to j}, \qquad \omega_{i\to j} \geq 0, \qquad \sum_{i\to j} \omega_{i\to j} = 1. \tag{5}$$

Equal weights correspond to no particular history and underlie the unweighted attenuation structure reported in Section 5.2. Weights set to the realized 1991–2013 national reallocation deliver the national experiment; and weights drawn from each state's own realized history deliver the state history experiment (Section 5.2). In short, the system is robust and can deliver results for any weights corresponding to any desired "experiment" so long as the shares sum to 1 and the changes in shares sum to 0. For example, one could examine the effect of accentuating or attenuating the historical share shift of the top quintile; we examine several such alternative hypotheticals in Section 5.2.

2.3 Identification

The model is meant to capture redistributive responses to purely distributional concerns. However, redistributive fiscal policy is also driven by factors which reflect the absolute state of the economy. Indeed, to a large degree, taxes and transfers are dedicated to providing insurance against adverse economic circumstances, such as old age. To the extent that these factors are associated with the market income distribution, they must enter into the controls Y. For instance, inequality is plausibly related to economic growth, and the level of income may independently affect taxes and transfers in any number of ways. Likewise, shifts in the market income distribution may be caused by an aging population or the onset of recession, developments which may mechanically trigger social insurance spending. The baseline model and all extensions therefore include four controls. Market income per capita captures the overall level of income. The share of the population over 65 proxies for insurance against old-age destitution, the cyclical component of the unemployment rate for the business cycle, and the poverty rate for privation. Several of these are themselves potentially endogenous. As indicated above, fixed effects and time dummies also enter the model, and are included in all the results reported below.[10] See Section 3 for more detailed definitions and the data sources used to construct each variable.

Obviously, the taxes and transfers that transform $\boldsymbol{M}$ into $\boldsymbol{D}$ shape work incentives and hence influence $\boldsymbol{M}$ itself. Thus, to account for reverse causality and isolate structural, exogenous changes to the market income distribution, we assume that the explanatory variables are endogenous (all except the share of the population over 65) and use instruments to identify the parameters.

Our instruments, which are tested in various combinations, fall into three categories: proxies for local labor-market shocks attributable to national or international forces; four-year (or deeper) lags of the endogenous variables; and forecasts of the endogenous variables built from initial values and national-level trends. The idea behind the first set of instruments is that national or international developments are plausibly independent of the tax and transfer policies of any U.S. state. If exposure to these shocks varies geographically, any measure of the local impacts should yield consistent estimates. We employ four instruments of this type in various combinations. The first, based on Bartik (1991), is a projection of state employment growth rates (net of working age population growth) based on state industry composition and national industry trends. The second, following Autor, Dorn, and Hanson (2013) and Asquith, Goswami, Neumark, and Rodriguez-Lopez (2019), measures import pressure by relating national imports by industry to each state's share of national industry employment. The third and fourth instruments, inspired by Kehrig and Ziebarth (2017), are interactions of the price of a globally traded commodity (or commodity index) with the share of state employment

[10]The only exception is for the out-of-sample tests, where we consider both the case with the time fixed effect from the most recent year and the case without the time fixed effect.

dedicated to its production.

The rationale for the second category is that the omitted variables which make up the error terms in (1) are essentially political in nature, tied to the regular election cycle, so that lags of four or more years plausibly precede the shocks that drive current redistribution. We confirm the validity of these lagged instruments directly with the difference-in-Hansen test reported in Section 7. Standard errors are clustered by state throughout. The last set of instruments is a substitute for the second set for the purpose of robustness exercises. The approach is taken from Voorheis, McCarty, and Shor (2015). For details on the variable definitions and data sources, see Section 3.

The instruments, like the model, presume that redistribution is shaped substantially by voters in the states themselves—through both their state officials and their federal representatives. Although federal taxes and transfers exceed those of any single state, state fiscal agency is large and well documented: states administer federal programs such as Medicaid and TANF (cash welfare) with wide discretion over benefit levels, eligibility, and total spending; they can buttress or offset national fiscal policy through their own taxes and transfers; and federal policy itself is set by senators and representatives accountable to state constituencies, whose delegations routinely act across party lines in their state's interest.[11] State-level choice is thus a substantial determinant of how market-income shifts pass through to disposable income.

A more specific concern is the exclusion restriction itself. Our labor-market instruments must affect the *disposable* distribution only through the *market* distribution they instrument. This is not automatic, because trade, commodity, and industry-demand shocks move local employment and earnings, and several transfer programs, unemployment insurance, SNAP, the EITC, and Trade Adjustment Assistance, respond to exactly such events. If a shock shifted disposable shares through a channel orthogonal to market shares, the restriction would fail. Three considerations lead us to discount this. First,

[11] See Weingast, Shepsle, and Johnsen (1981) and Clemens and Veuger (2021), who demonstrate the value of federal representation to states' allocation of federal funding.

the dominant programs respond to the *level and distribution of market income*, which are precisely the endogenous regressors the system conditions on. Two states with the same market-income distribution draw the same formula-driven transfers regardless of which shock produced that distribution, so the stabilizer response operates *through* the instrumented market shares rather than around them. Second, the one major program keyed to a shock's *cause* rather than to its income consequences, Trade Adjustment Assistance, triggered by import competition, is economically negligible in the disposable distribution, its outlays are orders of magnitude smaller than the unemployment-insurance and nutrition programs alongside it. Third, and most telling for our central result, any uncorrected mechanical-stabilizer channel would *inflate* measured protection of the low-income quintiles, since the means-tested programs at issue are bottom-weighted, yet we estimate that the bottom quintile receives the least protection. The threat thus biases *against* our main finding rather than producing it, and the overidentification and difference-in-Hansen tests of Section 7 do not reject the instruments' joint validity.

2.4 Estimation Procedure

We estimate the five-equation system (1) simultaneously by the Generalized Method of Moments (GMM), instrumenting the endogenous market income levels and controls with their own deep lags and the external shocks of Section 3.3. We use a two-step estimator with a heteroskedasticity-robust weight matrix and standard errors clustered by state. The share marginal effects of interest are then recovered from the estimated coefficients as described in Section 2.2. We also examine dramatically smaller instrument sets (Section 6), and confirm the system's identification with a full battery of econometric tests—the Hansen overidentification and difference-in-Hansen (C) tests, Sanderson–Windmeijer first-stage F-statistics against the Stock–Yogo thresholds, and weak-instrument-robust Anderson–Rubin inference—reported in full in Section 7.

Once the model is estimated, we compute the marginal effect at mean values of a one percentage point transfer of market income from the base (excluded) quintile to another (considering all four included quintiles in turn) on each of the disposable income shares. By construction, these marginal effects must sum to zero, since the disposable income shares always sum to 1. We note that, over our sample, all quintiles experience changes in shares of this magnitude.

The last step is to repeat the process for each base quintile. Simply interchanging the excluded quintile may be redundant in the sense that one set of estimates might be perfectly predictable from the other. Indeed, our baseline model imposes a symmetry condition. The marginal effect of a shift of market income share from one quintile to another is equal and opposite to that of the reverse transfer - along with a host of other cross-system restrictions (see Appendix A). The correspondence arises for the same reason that the choice of whether to include a dummy variable or its complement (e.g. a male or female indicator variables) is irrelevant in linear regression. However, all such restrictions may be relaxed by introducing nonlinearities or using different instruments for each equation. Thus, quadratic specifications provide a robustness check (see Section 6), and extensions which include interaction terms are also unconstrained.

More generally, none of these estimation choices is load-bearing. Section 6 shows that the middle-quintile attenuation peak and the near-zero bottom are unchanged by the instrument count (39, 25, or 22 instruments), by the price deflator (nominal or fully PCE- and RPP-deflated).

3 Data Sources, Sample, and Variable Definitions

We have delayed publication of our analysis, hoping to obtain consistent post-Affordable Care Act data on Medicaid by state by risk group, given the potentially significant effect the expansion had on the post-tax and transfer income distribution. That would also have been a natural test of any lagged or cumulative effect of inequality changes on redistribution. But data were at first unavailable, then delayed, then available only for eleven (unnamed states), then finally released for 2017-2018, but not for 2014, 2015 and 2016. Data on fungible values for Medicaid (described below) were discontinued. BLS added some additional state data, and Census corrected errors in its tax imputations, but we were unable to obtain information on which states had certain data imputed by which models. So we first present our analysis based on the most consistent, uninterrupted, data available. We discuss sensitivity and sample expansion, e.g. to additional state inclusions, various alternative measures for the value of government subsidized health insurance, and (in our view less reliable) results based on data through 2017, using the less accurate measure of Medicaid available, in Section 6 below. Beyond that date, unfortunately, the ASEC no longer includes data on the imputed income from owner occupied housing, a major source of income for a majority of households. Nor does it include data for the large and growing employer contributions for health insurance that cover a majority of the population. And finally it no longer includes data on property taxes.

Some states were dropped in our original analysis due to missing data for important instruments, such as BLS employment data by NAICS. Five of those states are added in the sensitivity analysis. Our interaction with BLS reveals that sometimes they use a statistical model to avoid problems with sample size that leads to the inability to produce a data series. But for privacy reasons, BLS has made a policy decision not to provide information to the public on which series are model based and which are sample based. Therefore, we include these five additional states only as a sensitivity check.

CPS stopped including Medicare/Medicaid data after the 2015 survey year. We explored the Kaiser Family Foundation (KFF), which provided information updated to 2014 based on analysis of the data from the 2014 Medicaid Statistical Information System and the Urban Institute's estimates from CMS-64 reports. Adjustments were made for 21 states because "spending per enrollee may not match the MSIS data or states own reporting systems."

The Centers for Medicare and Medicaid Services (CMS) finally released per capita spending by enrollment group by state for Medicare, but not for Medicaid. The 2019 Medicaid/CHIP scoreboard shows 2017 average Medicaid spending by state by enrollment group, but only for an incomplete list of states. As for the gaps for 2014/2015/2016, CMS stated "there are ... few Medicaid Public Use Files (PUF) and this very well could be due to state data submission."

3.1 Income Distribution Variables

All income distribution variables (market income shares, disposable income shares, market and disposable income Gini coefficients) are based on survey data from the ASEC March supplement of the Current Population Survey (CPS). The principal advantage of the CPS (and other survey data sources) is its scope. It contains information on both taxable and non-taxable sources of income, transfers, and the composition of families and households. Of course, researchers interested in measuring income inequality (especially top income shares) have often eschewed the CPS to avoid censored data, namely, income variables capped by a "top code" for privacy reasons. However, recent releases apply a "rank proximity swapping method" which reshuffles reported values in a way that obscures the identities of survey respondents while preserving the income distribution. What's more, the Census Bureau has released revisions based on current methods for survey years dating back to 1975. We utilize these corrections to construct a consistent data series.

Market income includes earnings, capital income, retirement/survivors'/disability pensions and annuities, private educational assistance, child support/alimony, the imputed return on home equity, and employer contributions to health plans, among other sources.[12]

[12]Capital income is composed of interest income, dividends, rents, royalties, estate and trust income.

Disposable income consists of market income plus transfers less taxes. Transfers include social security income, unemployment compensation, workers' compensation, veterans' payments, public educational assistance, cash welfare (TANF, etc.), disability (SSI), the (federal) earned income tax credit, food stamps (SNAP), school lunches, public housing and rental subsidies, low income energy assistance, and alternative measures, fungible and adjusted average spending, of the values of Medicaid and Medicare coverage, and employer provided health insurance,[13] among other sources.[14] Taxes include federal and state income taxes,[15] payroll taxes, and home property taxes but exclude corporate, sales, and excise taxes.[16] Although most variables are drawn directly from the survey, some are imputations provided directly by the Census Bureau.[17] These include all tax variables and most sources of in-kind income.

This income definition is fairly comprehensive. The only major omission is capital gains income, which is notoriously difficult to measure. The ASEC supplement actually includes imputations for realized capital gains, but only through 2008. As noted in Piketty and Saez (2003), realized capital gains are highly volatile and may reflect years of accrued income. Armour, Burkhauser, and Larrimore (2014) attempt to capture both realized and unrealized capital gains in each year, but their approach assumes that all assets of the same type yield the same returns. As in Piketty and Saez (2003), we choose

[13]Fungible values are computed by taking the minimum of the market value of the insurance policy and a family's disposable income after taxes and spending for basic needs (e.g. food and rent). We base values for medical insurance on Finkelstein, Hendren, and Luttmer (2019), 30% of average spending for Medicaid. We use 60% of risk group average state spending for Medicare and 80% for employer provided health insurance, but explore alternative values in Section 6. The value of employer provided health insurance has additional factors to consider, e.g. the tax subsidy, the fact that for most people marginal cost is a modest copay and (at least until the ACA) the higher cost on the individual private market.

[14]It is sometimes unclear in the data dictionaries whether a particular income source should be attributed to public assistance, public employment, or private employment. Wherever this is the case, we follow the methodology used in the Census Bureau's own P-60 research series.

[15]These tax variables do not account for credits. However, the data set contains a separate variable for the earned income tax credit.

[16]Accounting for the distributional effects of corporate, sales, and excise taxes would require estimates of substitution elasticities (across factors of production and consumer goods), tax incidence, and heterogeneity in consumer preferences across income percentiles. Hence, such measures would be highly speculative.

[17]Meyer, Wu, Finley, Langetieg, Medalia, Payne, and Plumley (2020) analyze the accuracy of tax imputations.

to exclude this source of income but would prefer to use a consistent, reliable series, were it available.

The unit of the analysis is the individual. However, given that households share both income and expenses, an individual's means and standard of living depend on the income and composition of the household. To account for this, we compute individual income by summing market and disposable income within households and then dividing the total by the 3-parameter equivalence scale of Betson (1996). Equivalence scales account for returns to scale in consumption. Expenses do not increase at the same rate as household size, children consume less than adults, etc. Among the many such scales that have been proposed, Betson's scale has been extensively analyzed in the poverty literature, is thought to best conform to the National Academy of Sciences recommendations for improved measurement, and yields inequality estimates in the middle of the pack.

The last step is to rank individuals according to equivalent income. It is then straightforward to compute income distribution statistics, from income shares to percentile ratios and Gini coefficients.

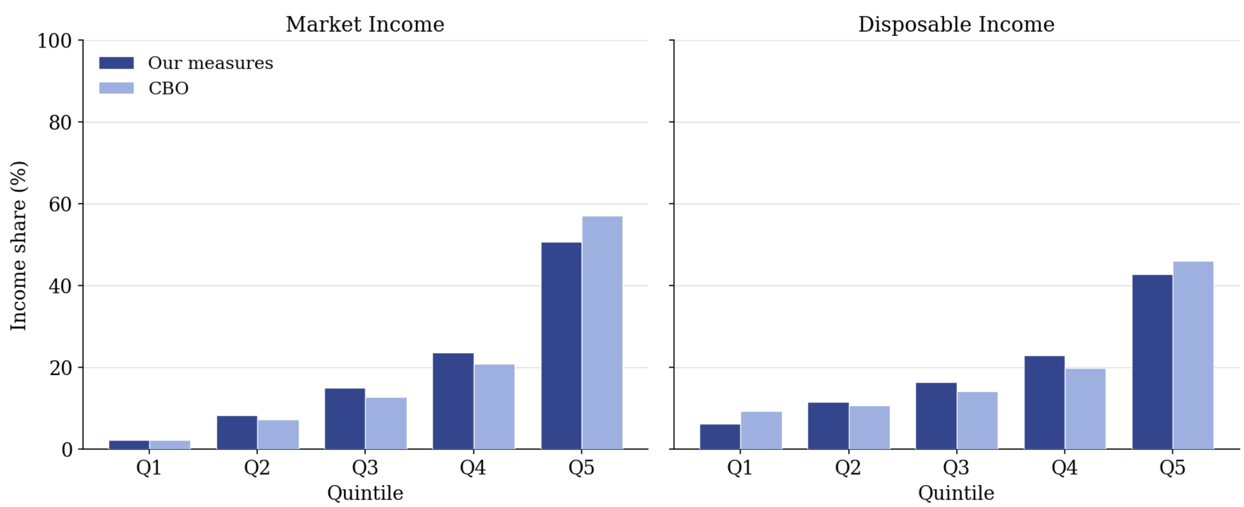


Figure 1: National Market and Disposable Income Shares by Quintile, 2013

Notes: This figure compares our constructed national income shares (2013) with those of Perese (2016), for market income (left) and disposable income (right). Dark bars are our measures. Light bars are CBO. The two are broadly consistent. Our measures show somewhat less concentration at the top and a smaller bottom-quintile disposable share, reflecting differences in income definition and imputation.

The resulting quintile shares (at the national level) are roughly consistent with the range of estimates reported in the literature. However, other researchers reasonably use different data sets and make different methodological choices, which can produce nontrivial differences in measured inequality. For comparison, Figure 1 depicts the U.S. distributions of market and disposable income in 2013 according to both our calculations and those of the CBO, and Appendix D provides a table comparing our definitions with those of some other leading studies (see Table A7).[18]

3.2 Controls

The baseline system controls for the overall level of income with market income per capita, built from the same CPS market-income aggregates as the quintile means. The poverty rate, the cyclical component of the unemployment rate, and the share of the population over 65 complete the controls and are described below.

For the real-income interaction of Section 5.2 and the fully-deflated robustness check of Section 6, we additionally construct real disposable income per capita by deflating a state's aggregate disposable income by its population and the local price level.[19] Aggregate disposable income is obtained from the Bureau of Economic Analysis (BEA), and population statistics are taken from the Census Bureau. Figures are converted from local contemporaneous dollars to 2008-national dollars using the national personal consumption expenditure deflator and state regional price parities (RPP). The BEA provides RPP for all 50 states (and D.C.) from 2008-2013. Imputations of RPP are used for the prior years.

[18]The CBO column in Table A7 is based on the definition used in Perese (2016) to calculate CBO's income distribution statistics for 2013, which we compare to our own 2013 numbers in Figure 1. Curiously, the CBO re-classifies Medicare and Social Security benefits as market income in a follow-up study for 2014. See Perese (2018). The most recent CBO release extends these estimates through 2021 (Congressional Budget Office, 2024), reporting a market-income Gini of 0.560 that transfers and federal taxes compress to 0.443.

[19]A more stringent deflator would also take into consideration heterogeneous consumption baskets among different income groups, which is interesting extensions of our analysis for future work.

RPP is imputed in two stages. First, we project RPP forward and backward from 2008 using estimates of relative state inflation. One estimate is based on the Federal Housing Finance Agency's (state-level) House Price Index, while another is based on the BEA's Gross State Product (GSP) price deflator series. The projections are scaled so that average state price levels (weighted by GSP) conform to national prices in every year. Next, we regress actual post-2008 RPP on both projections and then use the fitted regression model to impute values prior to 2008. In both the original and extended data sets, RPP ranges from about 0.8 to 1.15. Our findings are robust to the imputation procedure and the exclusion of RPP entirely. See footnote 29 in Section 5 for more detail.

Cyclical unemployment is computed by decomposing official (annual) state unemployment rates from the Bureau of Labor Statistics (BLS) into cyclical and structural components using a Hodrick-Prescott filter.[20]

Finally, we compute the share of the population over 65 using the ASEC supplement of the CPS and take the official poverty rate directly from the Census Bureau.[21]

3.3 Instruments

To construct our "Bartik national demand shock" instrument, we first compute predicted employment growth in each state-year as a weighted average of national employment growth by industry, where the weight on each industry is equal to its share of local (state) employment. The difference between these projections and working age population growth is then compounded year-over-year to capture the accrued effects on local labor market tightness. Employment by NAICS industry comes from the BLS, while population data is taken from the Census Bureau. We calculate working age population shares directly using the CPS ASEC supplement.

To construct our "import shock" instrument, gross national imports in a given in-

[20]The filter has parameter $\lambda = 1 \times 10^7$.

[21]We account for the fact that the poverty rate may be endogenous. See Section 3.3.

dustry are allocated to the states according to their share of national industry employment. This "lost business" is then aggregated across industries and scaled by gross state product (GSP). We use Census Bureau data on national imports by NAICS classification downloaded from trade.gov. As before, we obtain employment by NAICS industry from the BLS and GSP from the BEA.

The two commodity interactions are given by (oil&gas PPI)*(mining employment share) and (agricultural sector PPI)*(farming employment share), where PPI stands for producer price index.[22] We obtain producer prices from the Bureau of Labor Statistics and agricultural, mining, and total employment data from the BEA.

The last set of instruments consists of imputations of the endogenous variables based on movements in the corresponding national-level variables. For the poverty rate and cyclical unemployment, we perform a state-specific regression of the panel variable on the national-level variable and use fitted values. For mean market income by quintile as well as market income per capita, the procedure is similar, but the regressions are in logs and fitted values are exponentiated for the final result. Final instruments for the market income shares are then calculated by dividing the given quintile's imputed mean income by total imputed mean income (where the total is the sum across quintiles).

All four of these instruments, the Bartik employment shock and the import-exposure and commodity-price measures, are shift-share (Bartik) instruments, and the modern literature clarifies what identifies them. Goldsmith-Pinkham, Sorkin, and Swift (2020) show that a Bartik instrument is numerically equivalent to using the underlying exposure shares as instruments, so that identification through the employment-growth shock rests on the exogeneity of states' industry-employment shares with respect to unobserved determinants of redistribution. For the import-exposure and commodity-price instruments the more natural reading is the complementary one of Borusyak, Hull,

[22] In the NAICS system, mining (21) includes oil and gas extraction (211). Mining, except oil and gas (212), and support activities for mining (213). Empirically, mining employment consists predominantly of employment in oil and gas extraction.

and Jaravel (2022), in which identification comes from the national import and world commodity-price shocks being as good as randomly assigned while the exposure shares may themselves be endogenous. The two views place the exogeneity requirement on different objects, the shares in the first case, the shocks in the second, but both are plausible here, since a single U.S. state's tax and transfer policy cannot move national industry trends or world commodity prices.

3.4 Interaction Variables

In some of the regressions below, we interact the market income shares with the number of branches of state government controlled by the Democratic party. This variable is defined as follows: the governor's office counts as one branch, an independent in the governor's office counts as half a branch, chambers in bicameral legislatures count as half a branch, and ties count as half a chamber (in our sample, there is no case in which one party has a plurality but does not have a majority). To construct the variable, we use data on the party affiliation of state governors collected from National Governor's Association website.[23]

3.5 Sample Size

Because the income (and health) questions of the ASEC supplement were redesigned in 2015, our sample ends in 2013. In order to utilize the Census Bureau's imputations for in-kind income, we also only use CPS data after 1990. Finally, due to limitations in data availability, some variables are missing for particular states. However, we are able to construct a balanced panel data set containing all variables for 41 states and the

[23]See https://www.nga.org/cms/governors. Information on the control of state legislative chambers comes from the Statistical Abstract of the United States and the website of the National Conference of State Legislatures (U.S. Census Bureau, 2012). In some years, the number of seats held by each party in a given state is unavailable. These gaps are filled in by setting missing values in odd (even) years equal to the values in the subsequent (previous) year on the assumption that elections take place in even years. See http://www.ncsl.org/research/about-state-legislatures/legislator-data.aspx.

years 1991-2013. This truncated data set is the basis of most of our regressions. We do report in Section 6 analogous results using data updated through 2017 with less reliable post-Affordable Care Act Medicaid data mentioned above, and also extended to five additional states whose data are not perfectly consistent with the original 41 states.

4 Experiments of Interest

The National Picture

Once the model is estimated, it is possible to compute the marginal effect of a one percentage point transfer of market income share from one quintile to another on the disposable income shares of all quintiles. Clearly, any transformation of the market income distribution (by quintile) can be constructed as a sequence of such simple "experiments." For instance, between 1991 and 2013, the market income shares of quintiles 1 through 4 fell by 0.36, 1.51, 1.56, and 0.85 percentage points, respectively, and the top quintile gained 4.28 percentage points (pp) in market income share. Figure 2, discussed below, compares the model's predicted national disposable income distribution with the actual one. This history can be represented by the scenario

$$.36\,(Q_1 \rightarrow Q_5) + 1.51\,(Q_2 \rightarrow Q_5) + 1.56\,(Q_3 \rightarrow Q_5) + .85\,(Q_4 \rightarrow Q_5) \tag{6}$$

Of course, there are infinitely many alternative combinations of Q_i to Q_j transfers which net out to the same pattern of gains and losses (in market income share) across quintiles. However, given the nature of the model, the predicted effects are invariant to the particular construction or sequence of events (for a discussion of path dependence, asymmetries, etc., see Section 8).

As discussed in Section 2, there are 20 ordered quintile pairs and five effects for each experiment, for a total of 100 marginal effects, although only 50 are independent. Table

A3 provides the full set of GMM point estimates in tables organized by the "giving" quintile, assumed to be losing market share (i.e. the base quintile excluded from the regression). Each entry represents the marginal effect, at mean values, of a one percentage point shift of market income share from the base quintile to the row quintile on the disposable income share of the column quintile. For instance, the bottom row in the first table corresponds to the experiment $Q_1 \rightarrow Q_5$, and the -1.125 figure in the bottom left corner indicates that when the top quintile gains 1 pp of market income share at the expense of the bottom quintile, the disposable income share of Q_1 itself falls by 1.125 pp. In other words, the bottom quintile gets nothing back, if anything, its disposable income share falls by slightly more than the one point it surrendered in market income. Likewise, the 0.829 figure in the bottom right corner indicates that when $Q_1 \rightarrow Q_5$, the disposable income share of Q_5 itself rises by 0.829 pp. In a sense, the top quintile gives only 0.171 pp back after taxes and transfers. We shall refer to this dampening of shifts in market income as attenuation.

The construction in (6) makes concrete how any historical shift in the market distribution maps into the disposable distribution. Applying the estimated model to the national series in this way reproduces the 2013 national disposable income shares closely. The predicted shares for Q_1 through Q_5 are 6.3, 11.6, 16.5, 22.8, and 42.9 percentage points, against actual values of 6.2, 11.5, 16.5, 23.0, and 42.8 (Figure 2), an average absolute deviation well under a tenth of a percentage point. The formal out-of-sample exercise in Section 7, which predicts a held-out year's distribution from its market distribution alone, confirms the fit more demandingly.

A surprising pattern of disposable income share changes over the period arises because Q_5 retains substantially more than transferring quintiles Q_2 or Q_4 ultimately give up. This discrepancy can only be explained by losses for quintiles which neither gain nor lose market income share directly in the experiment. We shall term such effects spillovers. For instance, Q_3 is consistently affected when other quintiles transfer market

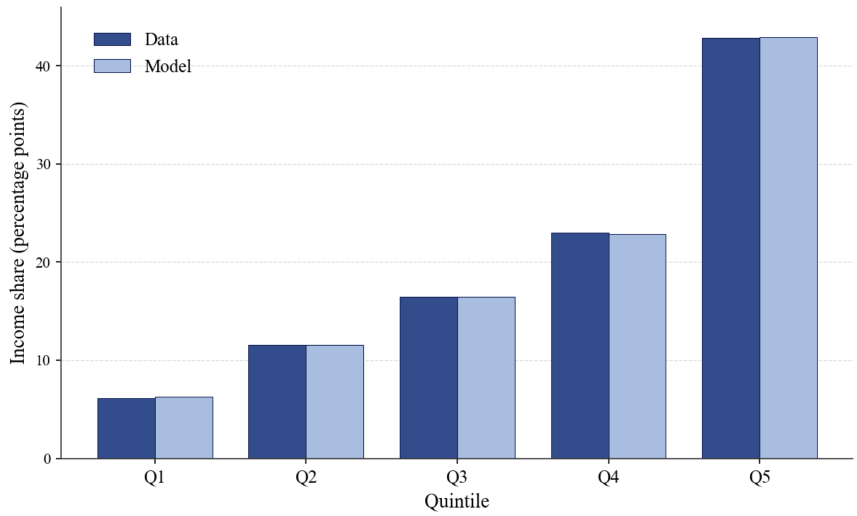


Figure 2: National Disposable Income Shares by Quintile, 2013: Data vs. Model

Notes: Actual (Data) versus model-predicted national disposable income shares by quintile in 2013. The model is estimated on the state panel, with state fixed effects removed and applied to the national series. Predicted and actual shares differ by less than a tenth of a percentage point on average.

income to Q_5, losing 0.090, 0.127, and 0.184 percentage points when $Q_1 \to Q_5$, $Q_2 \to Q_5$, and $Q_4 \to Q_5$, respectively. Q_1 also shares in losses. Its disposable income share falls by 0.105 and 0.340 percentage points when $Q_2 \to Q_5$ and $Q_4 \to Q_5$, respectively. We believe our study is the first to estimate large, statistically significant spillovers of this kind. Indeed, for the nation as a whole, such spillovers can be larger than the direct effects of tax-transfer adjustments.

As it happens, the shifts in the market income distribution over this period are largely representative of most state histories. For instance, the market income share of the top quintile increased in all but Minnesota. Likewise, the market income shares of Q_1, Q_2, and Q_3 fell in all but 7 states, 1 state, and 3 states, respectively. A table listing the profile of market income share changes by quintile for all 41 states is provided in Table A6.

It follows that the set of experiments $\{Q_i \to Q_5\}_{i=1}^4$ are of particular interest. Inspecting the GMM estimates in Table 1, we find that the top quintile keeps most of its gains in these experiments, *except* when such gains come at the expense of the middle quintile. For example, as previously indicated, Q_5 retains 0.829 pp when it gains 1 pp of market income share at the "expense" of Q_1. Similarly, when $Q_2 \to Q_5$ and $Q_4 \to Q_5$, the increases in Q_5's disposable income share are 0.972 pp and 1.393 pp, respectively. In contrast, when $Q_3 \rightarrow Q_5$, Q_5 ultimately keeps only 0.385 pp of its gain. Among the transferring quintiles (Q_1 through Q_4), the middle quintile gets the most back (ultimately losing only 0.360 pp), while the bottom quintile fares worst of all, losing 1.125 pp, even more than the one point it surrendered. Q_2 and Q_4 lie in between, losing 0.568 and 0.837 pp, respectively. As noted earlier, "spillover" effects (on quintiles which experience no change in their market income shares) are generally different from zero.

Table 1: Change in Disposable Income Distribution to a Change in Market Income Distribution

Quintile	$\Delta D1$	$\Delta D2$	$\Delta D3$	$\Delta D4$	$\Delta D5$
$M1 \rightarrow M5$	-1.125***	0.107	-0.090	0.279**	0.829***
$M2 \rightarrow M5$	-0.105	-0.568***	-0.127*	-0.173	0.972***
$M3 \rightarrow M5$	0.104	-0.118*	-0.360***	-0.011	0.385**
$M4 \rightarrow M5$	-0.340***	-0.031	-0.184***	-0.837***	1.393***

Notes: The table shows the marginal effects on disposable income shares when quintile i loses one percentage point of market income share to quintile j ($M_i \rightarrow M_j$). Estimates are from the baseline GMM specification. Significance: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

On the whole, the bottom quintile fares far worse, and the top quintile fares far better, than conventional wisdom might suggest. Consistent with the median voter hypothesis and other theories which postulate a special role for the middle class, Q_3 stands apart, both in the extent to which it is compensated as well as its effect on the top quintile. Finally, given Q_5's generally high retention rate and the prevalence of negative spillovers, we may conclude that taxes and transfers stabilize disposable income inequality more by spreading losses rather than redistributing gains.

For the most part, these insights depend critically on the use of instruments to account for endogeneity in the regressors. When the marginal effects are estimated by SUR (presented in Appendix B), Q_5 generally keeps less of its gains (around .75 pp), and the bottom three quintiles all get about .5 pp back when transferring 1 pp of market income to the top quintile. Q_4 is the exception, getting only about a quarter of its loss back, while the top quintile retains most of the transferred share. Thus, relative to GMM, the SUR estimates overstate the degree to which Q_1's losses and Q_5's gains are compensated and fail to capture the uniqueness of the middle quintile. This is the direction reverse causality predicts. Because redistribution and taxation themselves depress the market income of the quintiles they target, primarily through labor-supply and reporting responses, a quintile that is heavily compensated will, all else equal, show a lower market share, manufacturing a spurious positive association between transfers received and market income lost that an uninstrumented regression reads as high pass-through. The bias is largest where the policy response is largest, the bottom and lower-middle, which is exactly where the two estimators diverge. Instrumenting with external trade, commodity, and demand shocks isolates movements in market shares that are not themselves a response to redistribution, deflating the spurious bottom compensation and uncovering the sharp middle-quintile peak. See Table A1 for the full set of results. While we would usually from this point on present only our GMM results, we will on occasion below contrast them to analogous SUR results to illustrate the very different interpretation of the political economy mechanism consistent with each.

It is interesting to consider how the marginal effects of $\{Q_i \rightarrow Q_5\}_{i=1}^{4}$ depend on the levels of various interaction terms. Most have limited impact,[24] but past levels of inequality (the market income Gini coefficient at a 2-year lag) have meaningful effects. Higher past inequality flattens the middle-quintile attenuation peak and shifts protection toward Q_2, an effect we document more fully in Section 5.2.

The level of income has another potent effect. Although the marginal effects of $\{Q_i \rightarrow Q_5\}_{i=1}^{4}$ are largely robust to the other interaction terms, differences in real disposable income per capita reshape the redistribution geometry. The reshaping, however, falls mainly on the cross-quintile spillovers rather than the own pass-through. As real income rises, the bottom quintile becomes less protected (its attenuation falls), the middle quintile is essentially unaffected, and the spillovers borne by quintiles that neither

[24] Among the immaterial interactions is partisan control of state government. The estimates under full Democratic and full Republican control are nearly identical.

gain nor lose shrink, a sign of more narrowly targeted redistribution. We document this pattern in detail, with the supporting figure, in Section 5.2.

Taken together, the implications of the estimates are clear. It appears that, *in our sample*, the conventional wisdom that higher income societies redistribute more does not hold universally for all income groups. In fact, the opposite may well occur. For a full explanation, see Section 5, but a more advanced explanation focuses on the effect of the largest transfer program, Social Security, and on the fact that the highly redistributive state policies in states like New York and California, which have per-capita incomes that, when adjusted for price differences and household size, are not high.

A Broader Set of Experiments

The discussion so far has focused exclusively on the four experiments in which the market income share of the top quintile increases. Widening the scope of the analysis to the broader, but still historically relevant, set of experiments in which any quintile gains market income share at the expense of another quintile, $Q_i \to Q_j$, it is possible to explore more general questions. For instance, do higher quintiles fare worse, as might be expected under a progressive system of taxes and transfers, and more stringently, are these effects monotonic? We discuss transfers to higher quintiles, because as discussed below, the results are symmetric for transfers in either direction between any pair of quintiles.

To maintain comparability, this question must be subdivided into a series of more specific questions. First, fixing the giving quintile, do higher receiving quintiles fare worse, and does the giving/losing quintile do better as the receiving quintile's (percentile) rank increases? Examining the GMM estimates in Table A3, we see that for $\{Q_1 \to Q_i\}_{i>1}$ and $\{Q_2 \to Q_i\}_{i>2}$, there is no obvious monotonic pattern as hypothesized when Q_1 is losing. For instance, transfers of market income share from Q_1 to Q_2, Q_3, Q_4, and Q_5 lead to increases of 0.68, 0.27, 1.12, and 0.83 in the disposable income shares of the receiving quintiles, respectively. However, when Q_3 is the losing/giving quintile, Q_4 does in fact do better than Q_5, retaining 0.83 as opposed to 0.39 for the top quintile. As for the effect on the giving/losing quintile, there is no trend for Q_1. Transfers of market income share from Q_1 to Q_2, Q_3, Q_4, and Q_5 reduce Q_1's share of disposable income by 1.02, 1.23, 0.79, and 1.13, respectively. For Q_2 and Q_3, losses are monotonic but in the reverse direction.

Second, fixing the receiving quintile, do higher (percentile) giving quintiles get less back, and does the receiving quintile keep more as the giving/losing quintile's (percentile) rank increases? Again, there is either no trend, or the estimates are monotonic in reverse. As indicated above, when Q_5 is the receiving quintile, the bottom quintile Q_1 actually gets the least back (losing 1.13), while Q_3 gets the most back (ultimately losing only 0.36). For $\{Q_i \to Q_4\}$, Q_3 also does best, ultimately losing only 0.18 pp in disposable income share, whereas Q_1 and Q_2 lose around 0.79 and 0.54 respectively. When Q_3 is the receiving quintile, Q_2 does significantly better than Q_1, losing only 0.45 as compared to 1.23 for Q_1. As for the effect on the receiving quintile, Q_5 and Q_4 give back the most by far when receiving market income from Q_3 and Q_2, respectively (not when receiving from Q_1, as might be expected under a strict redistributionist mechanism), and Q_3's disposable income share rises more when $Q3$ gains market income at the expense of Q_1 than from Q_2. This suggests great sensitivity to fortunes of the middle quintiles.

Finally, do higher quintiles fare worse among those which neither gain nor lose market income share in the experiment? When market income shifts from Q_1 to another quintile, the spillover effects are not monotonic and are not uniformly signed. When $Q_1 \to Q_2$, Q_5 loses 0.14 and when $Q_1 \to Q_4$ it loses 0.56, yet when $Q_1 \to Q_3$ it gains 0.44. The spillovers are large but do not follow a simple rank ordering.

In sum, any hypothesized monotonicity does not hold. Otherwise, either higher quintiles fare better, or the middle quintile dominates.

5 A Political Economy Perspective

This section takes a more systematic, if more speculative, approach to the analysis of our model. Rather than identify especially salient or historically relevant experiments and explore them in great detail (as in Section 4), we instead summarize, distill, and aggregate the full set of estimates in various ways in order to make broad inferences about the apparent political economy of responses to market income distribution changes. Hence, generalizations based on unweighted averages of the marginal effects or joint hypothesis tests, etc. may not be an accurate representation of any likely change in the inequality of market income. Nevertheless, the results are interesting and potentially informative, both themselves and for what they imply about political mechanisms. And, importantly, our estimated marginal effects can be adapted to estimate the effect of *any* alternative hypothesized, or actual, changes in the distribution of market income by simply applying probability weights to any specific set or subset of changes, so long as the weights are non-negative and sum to one. We provide two such examples, spanning a range of market income distribution changes from gains at the top and losses in the remainder to losses at the top and gains below, in Section 6.

5.1 Hypothesis Tests

In Section 4 as well as the appendices, p-values for the point estimates (and statistical significance, as indicated by the star-markings) are calculated under the following null hypothesis. For any quintiles l and k,

$$\frac{\partial D^l}{\partial m\left(k\right)} = \begin{cases} 1 & l = k \\ -1 & l = \text{base} \\ 0 & \text{otherwise} \end{cases} \tag{7}$$

According to equation (7), the disposable income shares of receiving quintiles rise by one percentage point, the disposable income shares of transferring quintiles fall by one percentage point, and the disposable income shares of all other quintiles remain the same. In other words, shifts in the market income distribution pass through perfectly to the disposable income distribution. Since perhaps the simplest test of the model is whether it can distinguish disposable income share changes from market income share changes, this is a sensible benchmark.

More substantive tests are also possible. For instance, since redistributive responses are largely embedded in the design of fiscal systems, it may be interesting to consider whether and how closely the estimates adhere to some particular fixed policy formula. This can be tested by computing the automatic feedback effects of the given policy formula for each experiment and then setting the null hypotheses equal to these automatic effects. Statistically significant deviations from the null would then reflect either discretionary policy responses (adjustments to policy induced by shocks to inequality or other factors) or a misspecified benchmark (i.e. current law embeds a different policy formula).

For example, such a policy formula might simply take the form of a (possibly progressive) fixed net tax rate schedule, where "net" means taxes net of transfers.[25] This may be one way citizens, politicians, and scholars conceptualize our current system or preferred alternatives. Of course, the automatic feedback effects of such a policy would depend on the particular schedule enacted and the way in which the market income distribution evolves. Whatever the details, however, the "cross-effects" should be approximately zero. That is,

$$l \notin \{k, \text{base}\} \Rightarrow \frac{\partial D^l}{\partial m(k)} \approxeq 0 \tag{8}$$

[25]Net tax rates are calculated by subtracting transfers received from taxes paid and dividing by total income.

In words, under a fixed net tax schedule, if a quintile's market income share neither rises nor falls in the experiment, its disposable income share should be roughly unchanged. This can be shown formally, but the intuition is simple. Disposable income is market income after net taxes.[26] Therefore, if an experiment involves neither a change in a quintile's market income share nor its net tax rate, its disposable income share will be unchanged, ignoring small differences related to fluctuations in the average net tax rate across all households (i.e. the ratio of total disposable income to total market income, reflecting the overall burden of the tax/transfer system).

As for the rest of the effects, a full set of null hypotheses is possible with an additional assumption. As shown in Appendix E, if tax rates are not only constant over time but (perhaps unrealistically) uniform within quintiles, the automatic effect of the policy corresponds to

$$e_k^l = \begin{cases} 1 & l = k \\ \frac{-D^k}{D^l} & l = \text{base} \\ 0 & \text{otherwise} \end{cases} \tag{9}$$

where e_k^l is the elasticity of quintile l's disposable income share D^l with respect to quintile k's market income share $m(k)$, defined as $\frac{m(k)}{D^l}\frac{\partial D^l}{\partial m(k)}$.

Appendix C.2 provides estimates of the elasticities for all 20 possible experiments and reports p-values, for all 100 resulting estimates with respect to (9). The large majority differ significantly from this fixed-schedule benchmark, and the null hypothesis is jointly

[26] Let d_i, m_i, and τ_i be quintile i's ex ante disposable income share, market income share, and average net tax rate, respectively, so that

$$d_i = \frac{(1-\tau_i)m_i}{1-\bar{\tau}}.$$

Let d_i', m_i', and τ_i' be the corresponding values ex post. If quintile i neither gains nor loses market income share in the experiment and $\tau_i = \tau_i'$, then

$$d_i' = \frac{1-\bar{\tau}}{1-\bar{\tau}'}d_i$$

where $\bar{\tau} = \sum_i \tau_i m_i$ and $\bar{\tau}' = \sum_i \tau_i' m_i'$ are the weighted average tax rates ex ante and ex post, respectively. In most applications, $\frac{1-\bar{\tau}}{1-\bar{\tau}'}$ is approximately equal to 1.

rejected. The results provide clear evidence that a more complex process is at work. Either taxes and transfers do not conform to a simple fixed schedule in practice, or market income inequality changes induce discretionary redistributionist policy changes over time, which seems more than plausible.

5.2 Attenuation and Spillovers

Definitions

Each set of regressions produces 100 point estimates (5 alternative losing quintiles, times 4 different potentially gaining quintiles, times 5 affected quintiles). To help distill the results, we combine and synthesize the marginal effects in various ways and supplement each set of estimates with tables of summary statistics. Again, because the "experiments" are not all equally likely, these summary statistics should be taken with a grain of salt. Nevertheless, the results are noteworthy.

The first step is to record the benefits, wins, losses, attenuations and spillovers associated with each experiment. We then average these values over all 20 possible gain/loss pairs and report the results in summary tables.

The terms are most easily defined by considering an example. Suppose quintile 4 gains 1 pp of market income share at the expense of quintile 2. We can represent this shift of market income distribution as in Table 2. Suppose further that, according to the model, the effect on the disposable income shares are -0.2, -0.5, 0.1, 0.4, and 0.2 for Q1 through Q5.

Table 2: Market Income Share Changes, %

	Q1	Q2	Q3	Q4	Q5
Mkt		-1		1	
Disp	-0.2	-0.5	0.1	0.4	0.2

Notes: The table shows the hypothetical experiment when Q2 loses one percentage point of market income share to Q4. *Mkt* is short for market income.

Comparing disposable income share changes to market income share changes, it is clear that quintiles 2, 3, and 5 gain from the tax-transfer system, while quintiles 1 and 4 lose out. The benefits reaped by quintiles 1 through 5, measured by subtracting the "Mkt" vector from the "Disp" vector above, are -0.2, 0.5, 0.1, -0.6, and 0.2, respectively. If a quintile's benefit is positive and statistically significant, it is said to have won. Likewise, if a quintile's benefit is negative and statistically significant, it is said to have lost.

To the extent that redistribution is the outcome of a political contest, benefits, wins, and losses provide an easy way to score the results. However, the figures provide little insight into the underlying process. To understand the mechanics and uncover any patterns, we use attenuation and spillovers to deconstruct the transformation of market income share changes into disposable income share changes at a more generic level.

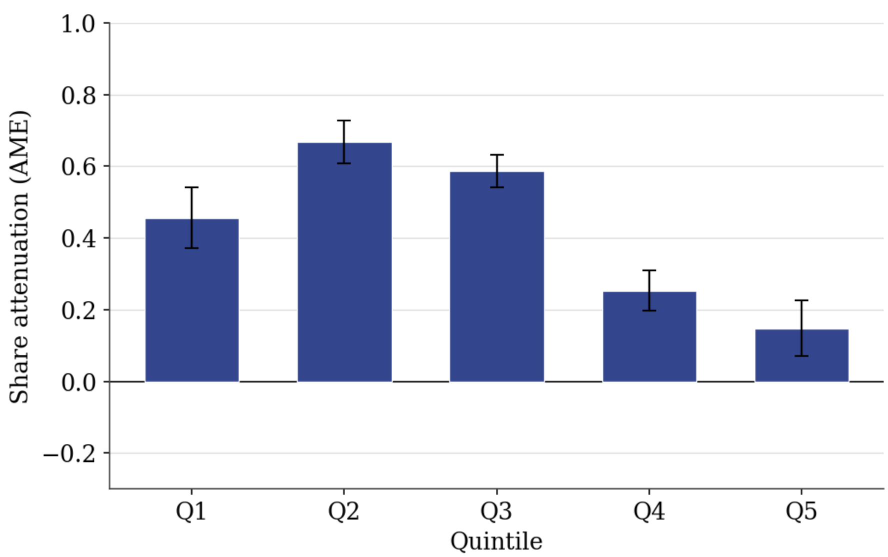
Attenuation denotes the degree to which initial (market) gains and losses are ultimately compensated. For instance, in the example above, Q_2 gets 0.5 pp of its initial 1 pp loss back, while Q_4 gives 0.6 pp of its initial 1 pp gain back. Thus, attenuation is 0.5 pp for Q_2 and 0.6 pp for Q_4. Because the estimates are highly symmetric, in the sense that the marginal effects $Q_i \to Q_j$ are approximately equal and opposite to that $Q_j \to Q_i$, the greater a quintile's attenuation on average, the more stable its share of disposable income. Thus, attenuation may be interpreted as a measure of the level of insurance provided by the tax-transfer system, the sensitivity of redistributive fiscal policy to a particular quintile's income share.

Spillovers are effects on quintiles whose market income shares are unchanged in the experiment. We attribute positive spillovers to receiving quintiles and negative spillovers to transferring/losing quintiles in the accounting. For example, in Table 2, 0.3 percentage points of Q_4's gains spill over to Q_3 and Q_5, while 0.2 percentage points of Q_2's losses spill over to Q_1. As emphasized in equation (8), spillovers are difficult to reconcile with static or fixed net tax schedules, so large spillovers provide evidence of more complex policy formulas or "active" policy responses.

Results

We begin our analyses with the SUR estimates, again given the stark contrast in potential political economy mechanisms with our GMM estimates. Figure 3 presents average attenuation by quintile under SUR estimation. Throughout, we refer to this diagram as the attenuation structure. Inspecting the figure, attenuation is high and broadly similar across the bottom three quintiles (in the range of roughly 0.5–0.7 pp, peaking at Q_2), before falling off dramatically for quintiles 4 and 5. In a sense, quintiles 1-3 are the most "protected" by redistributive fiscal policy. Thus, shifts in market income share are directly offset as much as they are scrambled and dispersed (indeed, gains and losses to Q_5 are barely shared).

Figure 3: The Attenuation Structure (SUR Estimation)

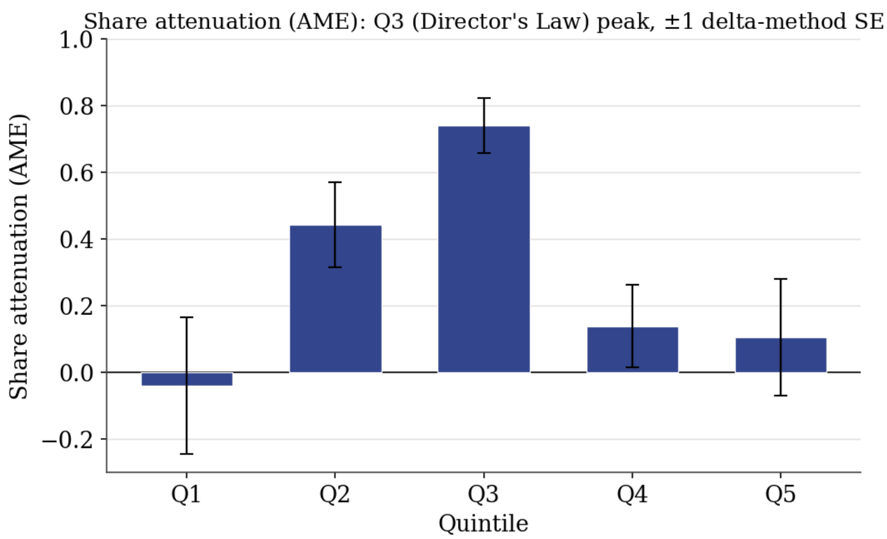


Notes: This figure presents average (recovered share) attenuation by quintile under SUR estimation, with delta-method standard errors. Attenuation is high across the bottom three quintiles (roughly 0.5–0.7 pp, peaking at Q_2), before falling off for quintiles 4 and 5. Quintiles 1-3 are the most "protected" by redistributive fiscal policy.

The results under GMM estimation (where we use the instruments to identify changes) are quite different. The attenuation structure depicted in Figure 4 reveals a striking parabolic shape in which attenuation peaks sharply at Q_3. The recovered share attenuation is -0.04, 0.44, 0.74, 0.14, and 0.11 for Q_1 through Q_5. The Q_3 peak is significant at

$p < .001$ ($z \approx 9$), while Q_1 is statistically indistinguishable from zero. Thus, redistributive fiscal policy is most sensitive to the fortunes of the middle quintile in particular (as opposed to the poorest quintiles more broadly). The sources of attenuation also differ. In contrast to the SUR results, spillovers are large. In other words, the government moderates market income share changes primarily by spreading gains and losses across the income spectrum. Marked disparities in the treatment of different quintiles are mostly due to spillover effects. We label this result the greedy median voter effect. This basic attenuation structure and the other qualitative results are robust to the inclusion of quadratic and interaction terms, and replacement of lagged values with imputations of the endogenous variables in the set of instruments.[27]

Figure 4: The Attenuation Structure (GMM Estimation)



Notes: This figure presents average (recovered share) attenuation by quintile under the baseline GMM estimation, with ±1 delta-method standard-error bars. The attenuation exhibits a parabolic shape peaking sharply at Q_3 (0.74, $p < .001$), with Q_1 near zero.

In essence, the contrast between Figures 3 and 4 illustrates the following: while the SUR estimates are roughly consistent with the conventional redistribution hypothesis, the GMM estimates are more in line with political economy median voter theory. Based on the SUR estimates, the government provides insurance to the poorest quintiles, and

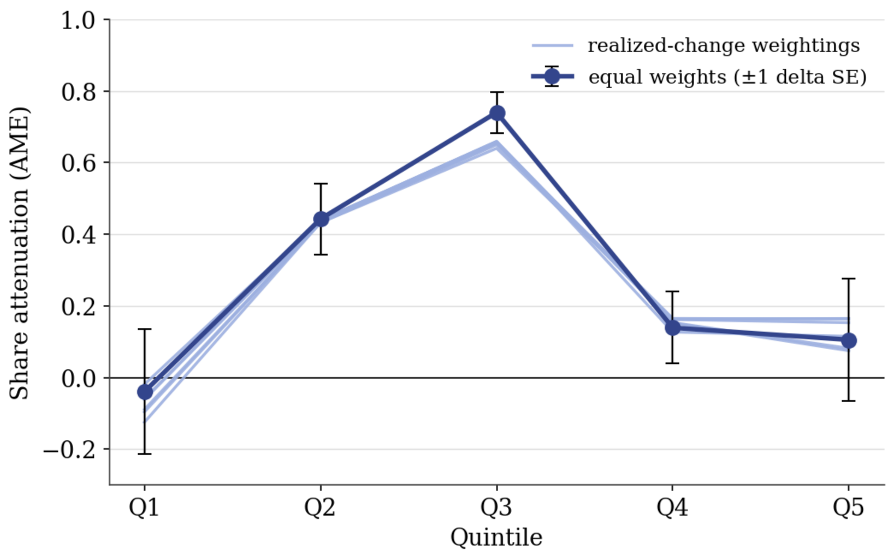
[27]The attenuation structure is also robust when the five additional states are added, although very slightly skewed towards the bottom with quadratic terms.

taxes and transfers counteract trends in the market income distribution. This accords with how redistribution policy is often understood or conceived. Using the GMM estimates, however, the middle quintile is more protected than any other. Changes in the income share of the middle quintile induce the greatest (compensatory) policy response. This is more consistent with the median voter hypothesis and what Stigler (1970) calls Director's Law. Redistributive politics favors the middle class at the expense of the rich and poor. At the same time, large spillovers indicate a high degree of flexibility in the political system/process, a broad feature of the voting and bargaining models in which inequality concerns shape legislation. Either policy is highly multidimensional and complex in design, or shifts in the market income distribution trigger "active" policy changes over time.

The structure in Figure 4 weights all 100 experiments equally, which corresponds to no particular history. Because the same estimates can be combined under any nonnegative weights summing to one, we can also map the *actual* historical reallocation of market income into its disposable consequences and ask whether the middle-quintile peak survives. It does. We re-weight by the realized 1991–2013 change, the familiar "bottom four to the top," in which Q_1 through Q_4 lost market share and Q_5 gained 4.28 percentage points, under six variants: the net endpoint difference (1991 versus 2013) and the pooled year-by-year flows (which capture the actual annual reallocations and within-period reversals rather than a single snapshot), each applied to the national distribution and to each state's own history (both population- and equal-weighted). Figure 5 overlays all six on the equal-weighted baseline. The middle-quintile peak is essentially invariant, Q_3 attenuation lies between 0.64 and 0.74 across every scheme, as are Q_2 (≈ 0.44) and Q_4 (≈ 0.15). What moves is the bottom and the top, and only with the endpoint-versus-year-by-year choice. The single-endpoint comparison, dominated by the secular bottom-to-top trend, makes Q_1 mildly negative (-0.13 nationally) and lifts Q_5 attenuation to 0.16, whereas the year-by-year version returns Q_1 to roughly zero (-0.02) and Q_5

to 0.11. We treat the year-by-year version as primary, since it reflects the actual annual dynamics rather than two endpoints. Either way the bottom quintile is statistically indistinguishable from zero, and the national and state-history panels are nearly identical, so the Director's-Law structure is not an artifact of how the experiments are weighted: it holds under the actual historical reallocation, whether we use the national aggregate or weight each state by its own realized experience.

Figure 5: The Attenuation Structure, National Picture



Notes: Recovered share attenuation by quintile under the equal-weighted baseline (navy, with ±1 delta-method standard errors) and under six realized-change weightings (light). The net 1991–2013 endpoint difference and the pooled year-by-year flows, each applied to the national distribution and to each state's own history (both population- and equal-weighted). The Q_3 peak is present and sharply significant under every scheme. Only Q_1 and Q_5 vary, and only with the endpoint-versus-year-by-year choice.

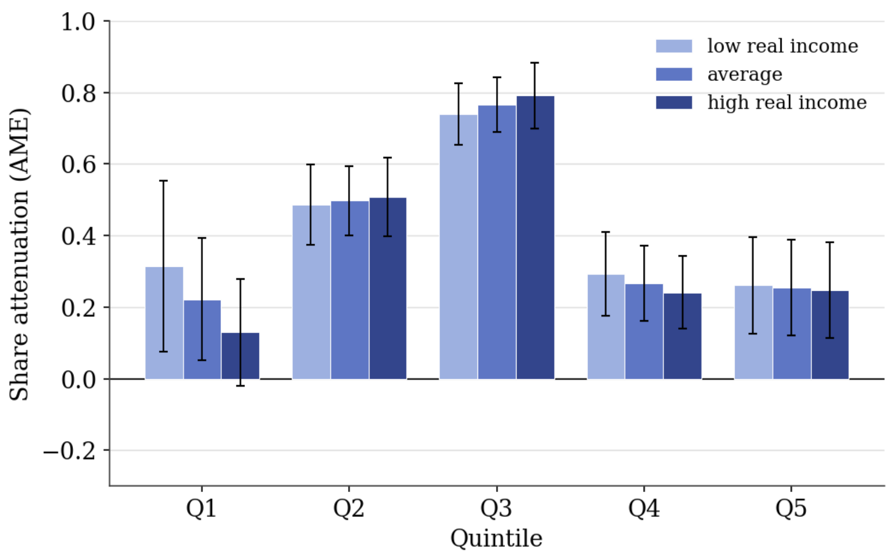
It is interesting to consider how these results vary with economic conditions. To supplement the analysis, we separately interact lagged market income Gini coefficients and real disposable income per capita with the market income shares and estimate the marginal effects at different points of evaluation.[28]

The attenuation structure is otherwise remarkably stable. The most important excep-

[28]Like most of the literature, we use the Gini coefficient as our measure of spread. There are alternatives, such as the Atkinson and Theil indices, but any discrepancies in measured levels of inequality typically do not carry over to measured changes (more relevant to this analysis). This is evident in data provided by the Census Bureau's annual income and poverty (P-60) reports, which contain a comprehensive set of inequality measures for the United States going back to 1967. See Semega, Fontenot, Kollar, and Kollar (2017), Table A-2.

tion is real disposable income per capita, which has sizable effects, though these operate mainly on the cross-quintile spillovers rather than on the own pass-through. Figure 6 illustrates the evolution of the attenuation structure. As average disposable income per capita rises, attenuation falls for the bottom quintile (from about 0.31 to 0.13 as real income moves from one standard deviation below the mean to one above), rises slightly for Q_2, rises modestly for Q_3, falls modestly for Q_4, and is essentially unchanged for the top quintile. In our sample, then, it appears that richer societies provide less insurance for the bottom quintile, not more.[29] This is consistent with the notion that the absolute, rather than relative, level of income for lower-income households (Q_1) is the primary source of voter concern.

Figure 6: The Effect of Per Capita Income on Redistribution



Notes: This figure shows the attenuation structure interacted with real disposable income per capita, evaluated at its −1 SD, mean, and +1 SD (recovered share attenuation, ±1 delta-method SE). As real income rises, attenuation falls for the bottom quintile, rises slightly for Q_2 and modestly for Q_3, falls modestly for Q_4, and is essentially unchanged for the top quintile.

We conjecture that this considerable inverse relation between redistribution and the

[29] This direction, and the basic shape of the attenuation structure, is robust to how real income is defined. Using a simple imputation procedure which substitutes the 2008 BEA relative price parities for all prior years, or dropping RPP entirely, the bottom quintiles still see declines in attenuation as real income increases. The income effect is sharply identified on the spillover (off-diagonal) cells rather than on the own pass-through, so it is best read as reshaping how redistribution is spread across quintiles rather than as a precisely estimated change in any single quintile's own attenuation.

level of real income may be due to the political pressure reflecting voters' greater concern with the level of low income than inequality per se. This result may seem surprising because, according to conventional wisdom, higher income states like New York and California tend to tax their own residents more and provide more state funding to safety net programs. However, this pattern is not apparent in the data. For instance, rankings of U.S. states based on real disposable income per capita, on the one hand, and state and local tax burdens, on the other, have a correlation of only 0.1. Misconceptions may stem from a failure to recognize that nominal incomes are a poor guide to real incomes. After adjusting for the cost of living, household size, and higher effective federal tax rates (federal income tax rates are applied to income that is not spatially indexed), California and New York actually lie on the bottom half of the income distribution. Indeed, California's high housing costs make it the state with by far the largest fraction of the population, almost one in five, living in poverty (Fox, 2017).

In any case, our results are not inconsistent with this hypothesis. Even if higher income states do, in fact, redistribute more, to the extent that this reflects different preferences for redistribution (due to culture, the red-state/blue-state divide, etc.), this cross-sectional variation is already captured by state fixed-effects. Moreover, our results concern overall or net redistribution, not state policy alone. As noted above, federal income taxes are applied to tax bases that do not reflect geographic cost of living differences, and the same is true for the bulk of transfer income, e.g. social security benefits. To a large degree, local taxes and transfers may be a response to the effect (or lack thereof) of federal programs. Indeed, the impact of federal policy varies considerably across states. Table 3 reports the results of a regression of the federal transfer-income ratio (total federal transfers over total disposable income in the given state-year) on relative state income, where relative state income is defined as real disposable income per capita, scaled by the average (across states) in the given year.[30] Since federal transfers

[30]Data on total transfers and total disposable income by state come from the BEA, while variables for state government spending on social insurance and "public assistance and subsidies" are taken from the

are largely directed to senior citizens (through Social Security and Medicare), we control for the share of the population over 65. Still, the effect is about -0.12 and significant at the 1% level. This means that for every 10% increase in per capita income (relative to the mean), the share of federal transfers in state income falls by 1.2%. Given that relative state income ranges from .80 to 1.50 in the sample and the mean federal transfer-income ratio is .171, this is quite a large effect. Thus, to the extent that higher income states do, in fact, have more generous transfer programs or more progressive tax codes, this may only partially compensate for less federal redistribution.

Table 3: Relative State Income v. Federal Share of Transfer Spending

VARIABLES	(1) Federal Transfers (Share of Income)	(2) Federal Transfers (Share of Income)	(3) Federal Transfers (Share of Income)
Relative State Income	-0.131***	-0.108***	-0.119***
	(0.011)	(0.009)	(0.008)
Population over 65		0.992***	0.811***
		(0.051)	(0.047)
Constant	0.305***	0.167***	0.199***
	(0.011)	(0.012)	(0.011)
Observations	943	943	943
R^2	0.141	0.360	0.532
Year FE	NO	NO	YES

Notes: This table reports the results of a regression of the federal transfer-income ratio (total federal transfers over total disposable income in the given state-year) on relative state income, where relative state income is defined as real disposable income per capita, scaled by the average across states in the given year. Robust standard errors in parentheses. Significance: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

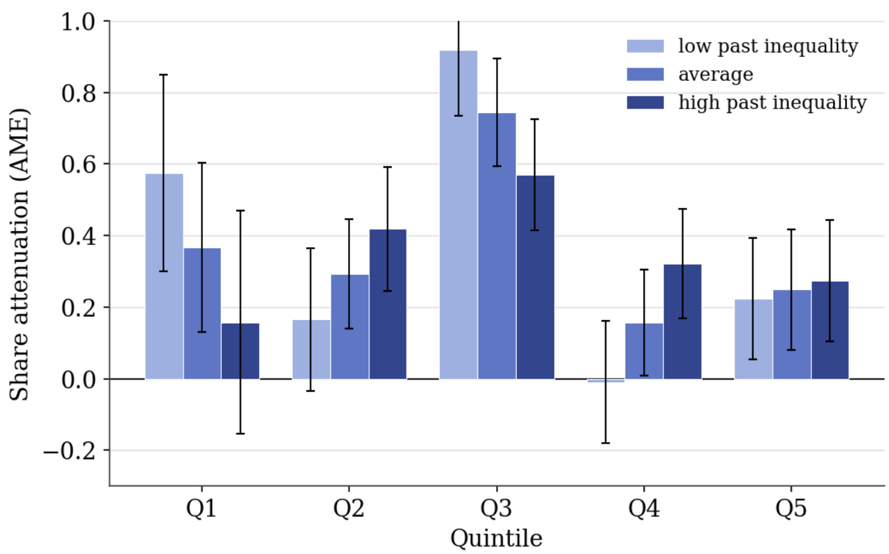
This negative effect of real income growth is precisely estimated for our sample of U.S. states over our sample period of recent decades. While the real income range is substantial, the low end, the poorest states at the beginning of the sample period, is still a quite high real income by international and historical standards.

Turning to the remaining interaction terms, we use both 2 and 4-year lags of the market income Gini coefficient as proxies for inequality trends. Here, past inequality has a meaningful effect on the attenuation structure. As past market inequality rises (from one

Census Bureau.

standard deviation below the mean to one above), the middle-quintile peak flattens and protection shifts toward Q_2. At a two-year lag, a specification that converges under two-step GMM on the shorter sample the two-year lag permits and comfortably passes the overidentification test (Hansen $p = 0.12$), bottom-quintile attenuation falls from about 0.58 to 0.16, the Q_3 peak falls from 0.92 to 0.57, and Q_2 rises from 0.17 to 0.42 (Figure 7). The four-year lag is directionally similar, though that longer-lag specification does not pass the overidentification test, so we rest the result on the two-year lag. In other words, higher past inequality erodes both bottom-quintile protection and the sharpness of the middle-quintile peak, tilting redistribution toward the lower-middle of the distribution. We read this as evidence that the history of inequality conditions the political response, not merely its contemporaneous level.

Figure 7: Attenuation by the History of Market-Income Inequality



Notes: Recovered share attenuation by quintile from the GMM specification that interacts the market shares with the market-income Gini coefficient at a two-year lag, evaluated at low (one standard deviation below the mean), average, and high (one standard deviation above) past inequality. As past inequality rises, the sharp Q_3 peak (0.92 at low past inequality) flattens to 0.57 and protection spreads toward Q_2 and Q_4. Past inequality significantly shifts own pass-through in 9 of the 20 own-quintile cells (11 at the 10% level).

What is driving our results?

As we have seen, several important facts and political economy inferences emerge from our results. Perhaps most intriguing is what we have labeled the greedy median voter. It appears the middle quintile does uniquely well in several dimensions. It gets back, via redistribution, the largest fraction of its loss. And it keeps the largest fraction of its gains in market income, after netting out any redistribution. And it does best with the redistribution spillovers when shifts in the market income distribution occurs between other quintiles.

Another interesting result is the sizable real income effect. As real per capita income rises, redistribution declines for the bottom quintile, working mainly through the cross-quintile spillovers, while the middle-quintile peak is little changed (see Figure 6). There are many potential explanations, but one that makes sense to us as a partial explanation is that voters have an absolute, not relative, notion of how redistribution should be targeted. So even as considerable inequality remains everyone, including the poor, gets richer, programs targeted at absolute poverty relief rather than tightening income inequality, are more prevalent.

Finally, the difference in attenuation structures between our GMM instrumented results and the estimates from our regressions lacking instrumentation is striking. Not only is the shape sharply parabolic with instrumentation, but in fact it shifts away from heavy redistribution to the bottom two quintiles at the expense of the top two (compare Figures 3 and 4).

While there are undoubtedly many competing or complementary explanations, the one that appeals to us is Director's Law (Stigler, 1970), that redistribution always favors the middle class.[31] In short, a political economy mechanism whereby the middle class

[31] Direct empirical tests of Director's Law are scarce. The main exception is Pamp and Mohl (2009), who, using Luxembourg Income Study data on income-decile share gains across 22 (largely European) OECD countries over 1971–2005, find that rising inequality redistributes toward the middle class at the expense of both the top and the bottom. Xu and Zou (2000) also find suggestive, though statistically weak, signs of it in the evolution of China's provincial income distribution.

is politically powerful and votes to enact policies for its benefit, even when the putative purpose, as FDR stated for Social Security, is "to secure a measure of protection against poverty ridden old age." Because we control for the share of the population over 65, this centering is not merely an artifact of where the elderly, and the Social Security and Medicare dollars they draw, happen to be concentrated. What remains, that the largest such program is itself middle-weighted by design, makes its incidence and the median-voter interpretation point the same way rather than compete. Either way the middle quintile is uniquely well served, through programs that voters have historically chosen to enact and sustain.

And Social Security is a main driver of this result. It is the largest federal program. It is by far the most important poverty reducer (Meyer and Sullivan, 2023) and the large majority of the benefits go to the broad middle of the income distribution. Furthermore, Social Security benefits are not spatially indexed so the same Social Security benefits to a retiree in New York or San Francisco and Alabama are transferring resources to somewhat higher up the real income distribution (adjusted for cost of living differences) in less expensive Alabama than in expensive New York and California. Also important is the underappreciated fact that the benefit formula used to calculate Social Security benefits generates rising real benefits over time. While once the beneficiary starts receiving benefits they are indexed by a measure of CPI inflation, the starting level of benefits are initially wage indexed. Despite the 2021-23 period when wage growth lagged behind the high inflation usually wage growth exceeds price growth, so over the quarter century covered in our sample real benefits typically grew by 20-25% for the latest vs. the earliest cohort of retirees.[32]

[32]This is independent of whether changes in the CPI may have overstated consumer inflation in this period (Boskin, Dulberger, Gordon, Griliches, and Jorgenson, 1997; Moulton, 2018).

5.3 Insurance Coalitions

If the marginal effects on two quintiles share the same sign and each is statistically significant relative to the mechanical lump-sum benchmark in (7), the two are said to form an insurance coalition. The notion is apt because reciprocal transfers of market income produce equal and opposite marginal effects: if the disposable income shares of two quintiles increase when, say, $Q_i \to Q_j$, they decrease when $Q_j \to Q_i$. Members of an insurance coalition thus share both gains and losses, stabilizing their disposable income shares. In the example of Section 5.2, the two insurance coalitions are $\{Q_2, Q_3, Q_5\}$ and $\{Q_1, Q_4\}$. For each estimator we record, over the twenty experiments, how often each pair of quintiles co-moves, reporting the frequencies in the matrices that accompany the estimates.[33]

Both estimators place the middle quintile at the center of the insurance structure. Under the GMM estimates in Table A3, Q_3 has the highest aggregate co-membership and belongs to both of the most frequent pairings, $\{Q_2, Q_3\}$ and $\{Q_3, Q_5\}$, each arising in six of the twenty experiments. The bottom quintile is the most peripheral, co-insured less than half as often. The SUR estimates (Appendix B) tell the same story, Q_3 is again the most frequent member, paired most often with Q_1 and Q_2, the difference being that under SUR it is the affluent quintiles, rather than the bottom, that are least integrated.

The two estimators differ revealingly in where they place the bottom. Under SUR the lowest quintile is itself well integrated, co-insured with Q_3 as often as any pair in the matrix, so that protection extends toward the bottom. Under GMM, by contrast, Q_1 is conspicuously isolated: it is the least co-insured quintile, while the middle's disposable share co-moves with both its lower neighbor and the top. On the instrumented estimates we prefer, then, it is the bottom rather than the top that lacks effective political protec-

[33]The experiments are unweighted and not equally likely in historical experience, so the frequencies are best read as ex-ante indicators of which quintiles' fortunes tend to co-move, not as forecasts. Broad blocs in which three or more quintiles gain simultaneously relative to the mechanical benchmark essentially never arise—in only one of the twenty GMM experiments and two under SUR—so a crude majoritarian, expropriate-the-rest reading finds no support here.

tion, the reading most consistent with the greedy-median-voter mechanism—the pivotal middle aligning to protect itself rather than the poor, as Iversen and Soskice (2006) emphasize in explaining cross-national differences in redistribution. The bottom quintile's isolation echoes Bonica, McCarty, Poole, and Rosenthal (2013), who tie the failure of U.S. democracy to arrest rising inequality partly to the low turnout and weak representation of low-income voters. That the middle quintile anchors the insurance structure under either estimator reinforces the median-voter reading of the attenuation pattern in Section 5.2.

6 Robustness

The preceding sections established our central results. Once we instrument, the attenuation of market-income shifts is hump-shaped in quintile rank, peaks sharply at the middle quintile, and is driven largely by cross-quintile spillovers. We now ask whether that result survives the concerns a skeptical reader would naturally raise. It does, along every dimension we can vary. The sharp middle-quintile peak and the near-zero bottom-quintile attenuation reappear in every perturbation we have tried, and the overidentification test keeps passing throughout. Figure 8 overlays the instrument-count, sample-size, and deflator variants, which trace essentially the same parabola. We take the underlying concerns in turn.

Are the results an artifact of using too many instruments? A very close in-sample fit can signal an over-fitting, too-many-instruments problem. Two pieces of evidence argue against it. First, the model forecasts the disposable distribution well *out* of sample (Section 7), so it is not merely interpolating the estimation data. Second, and most directly, we cut the instrument set from our baseline of 39 down to a Roodman-conservative 25 (Roodman, 2009), and to a still-leaner 22 that drops the aggregate income anchor's own lags. The middle-quintile peak and the near-zero bottom-quintile attenuation are unchanged. The Q_3 attenuation stays near 0.7 (it reads 0.74, 0.64, and 0.68 as the instrument count falls from 39 to 25 to 22) and Q_1 remains at zero, while the overidentification test continues to pass comfortably ($p = 0.17$, 0.17, and 0.15). What falls is only the count of individually significant dollar-level marginal effects, from about 76 to about 60 of 100, exactly as expected with fewer moments (the recovered *share* effects, of which 68 are significant in the baseline, behave the same way). The *shape* of the attenuation structure, the object we interpret, is invariant to the instrument count.

Are they specific to this sample of states or this period? Our main analysis runs over 1991–2013, the last year for which the CPS supports our full income definition. The detailed in-kind valuation of Medicaid and Medicare is discontinued afterward, and imputed rent on owner-occupied housing, employer health contributions, and property taxes disappear after 2017 (Section 3). We deliberately stop there rather than extend the panel onto degraded data. Within this window the result is unmoved by the natural perturbations. Adding the five states originally dropped for missing instrument data (46 states in all) leaves the Director's-Law peak intact (Q_3 attenuation ≈ 0.75, Hansen $p = 0.39$), as do the greedy-median-voter spillovers and the sharp contrast between the instrumented and uninstrumented estimates. We have nonetheless re-estimated year by year through 2017 on the less-reliable post-2013 data. We do not rely on those windows, but every quintile in them remains within the headline's 95% confidence interval, so even the degraded extensions leave the structure standing.

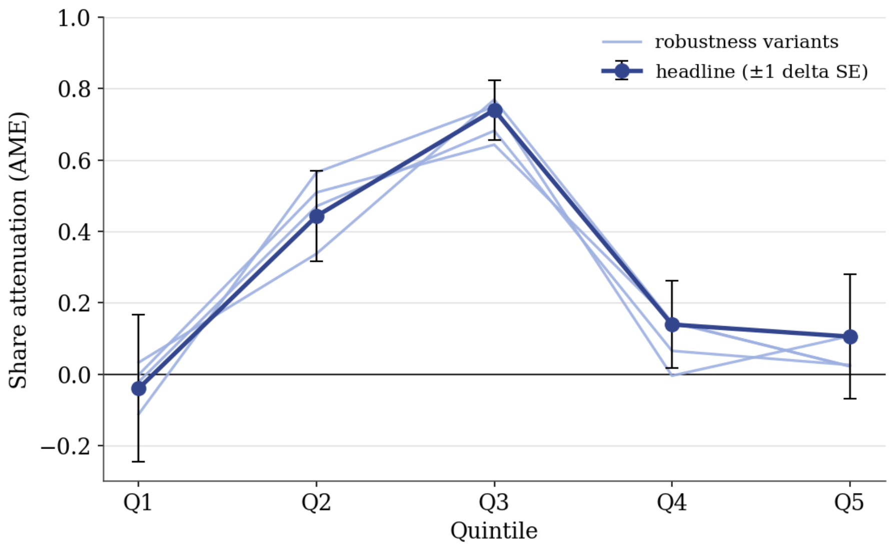
Are they an artifact of the nominal income units? The headline panel uses raw nominal income. We deflate neither for inflation over time nor for price differences across states. Any such deflation would be common to all quintiles within a state-year, so it cancels in every share, and hence in every marginal effect we report, and could touch the estimates only through the per-capita income level and the instruments, which are denominated in those units. As a direct check we make income fully real, applying both the national PCE deflator over time and the BEA regional price parities across space (with the pre-2008 parities imputed as described in Section 3). The share-attenuation profile is essentially unchanged, the middle-quintile peak is, if anything, marginally sharper (Q_3 attenuation 0.77). The structure is thus not an artifact of the nominal units. The most exacting deflation would go further still, letting the consumption basket, and thus the price index, vary across income groups, since inflation is not distributionally neutral (Carloni, 2025). We regard incorporating such group-specific indices as a natural refinement for future work.

Are they sensitive to the estimator? The two-step efficient estimator converges, and reports the overidentification test, for the baseline and for the two-year inequality interaction. Several of the interaction specifications, the four-year inequality lag, real income, and the party-control interactions, instead reach the iteration cap without converging, because the weak and collinear interaction moments make the two-step efficient weight matrix near-singular. For these we also estimate one-step GMM on the identical instrument set, which converges in a handful of iterations. The one-step and capped two-step estimates agree closely and always share the same peak quintile. The conclusions we draw from those interactions are therefore not artifacts of non-convergence in a handful of sub-scenarios we consider.

Are they sensitive to how in-kind income is valued? Our income definition imputes the value of Medicaid, Medicare, and employer-provided health insurance, each of which requires a valuation choice. Re-running with the Census "fungible" valuation, the budgetary resources a household frees up beyond food and housing, and with alternative spending adjustments leaves the qualitative results intact, most importantly the parabolic peak and the sizable real-income effect, with only minor movement in the measured *level* of inequality (a larger imputed value lowers measured inequality, consistent with Burkhauser, Larrimore, and Simon (2013)).

Are they sensitive to the model form? Finally, allowing the errors to be fully serially correlated, dropping the lagged instruments, partialing out the fixed effects by a Frisch–Waugh–Lovell procedure, and instrumenting instead with national-trend imputations in the manner of Voorheis, McCarty, and Shor (2015), with state-clustered standard errors, leaves the attenuation structure little changed. Quadratic specifications and alternative income definitions are likewise qualitatively similar.

Figure 8: Share Attenuation Across the Robustness Basket

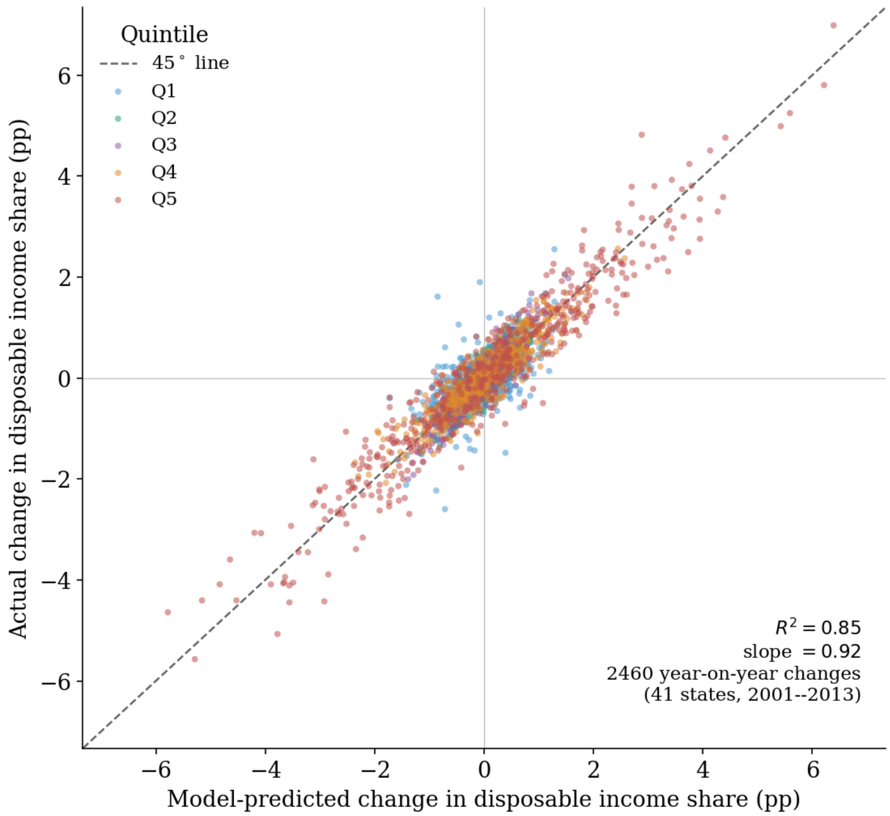


Notes: Recovered share attenuation by quintile for the headline specification (navy, with ±1 delta-method standard errors) and for each robustness variant (light). Instrument count (39 vs. 25 vs. 22 instruments), sample size (41 vs. 46 states), and the fully real (PCE + RPP) series. Every variant traces the same middle-quintile peak, with all five quintiles falling inside the headline's confidence band.

7 Model Fit and Instrument Validity

Reproducing the level of the disposable distribution is almost mechanical. Disposable and market shares are so highly correlated that any sensible model fits the levels to a pooled R^2 near one. The demanding test is whether the model tracks how the distribution moves. Figure 9 therefore plots, for every state and quintile, the model's predicted year-on-year change in the disposable income share against the actual change (41 states, 2001–2013, 2,460 changes). The model passes. The changes line up along the 45° line with a slope of 0.92 and an R^2 of 0.85, centered on the origin within every quintile, so there is no systematic over- or under-prediction.

Figure 9: Actual vs. Model-Predicted Year-on-Year Changes in Disposable Income Shares



Notes: Each point is one state-year-quintile change in the estimation sample (41 states, 2001–2013, 2,460 year-on-year changes). The horizontal axis is the model-predicted change in the disposable income share, the vertical axis the actual change, colored by quintile, with the 45° line dashed. The changes cluster tightly along the line (slope 0.92, $R^2 = 0.85$) and are centered on the origin within each quintile, so the model tracks how the disposable distribution moves, not merely its level.

The model also predicts well *out* of sample, which guards against over-fitting. For instance, removing state fixed effects and applying the model estimated using 1991-2012 to national data, the forecasts for 2013 are almost an exact match of the actual disposable income shares (see Table 4). In addition, to test whether the model can produce a good fit for the more recent trend of income inequality at the bottom, we use the estimates from 1991-2013 sample and compare its out-of-sample predictions for 2014 disposable income distribution to the data. The model predicted disposable income shares are consistent with the actual observed distribution.

Table 4: National Disposable Income Shares, 2013

	Q1	Q2	Q3	Q4	Q5
Actual	6.16	11.54	16.47	23.01	42.82
Predicted	6.25	11.56	16.46	22.83	42.91

Notes: This table reports the results of out-of-sample prediction when the model is estimated using 1991-2012 data before predicting the disposable income shares in 2013. *Actual* indicates the disposable income shares in the data, and *Predicted* values are from the model.

The near-perfect fit in *levels* could still prompt an over-fitting worry, but Section 6 confirms that paring the instrument set back from 39 to 25 or 22 instruments leaves the attenuation structure intact, so the fit does not reflect a too-many-instruments problem.

This predictive power does not come at the expense of precision, and the baseline clears the standard GMM diagnostics cleanly, on conservative settings throughout: a two-step estimator with a heteroskedasticity-robust weight matrix and standard errors clustered by state. Of the 100 recovered share marginal effects, 68 are significant at the 10% level (58 at the 5% level and 40 at the 1% level), and the mechanical null (7) is jointly rejected. Identification is equally clean. The Hansen test (based on the J-statistic) comfortably fails to reject the joint exogeneity of the instruments, with a p-value of about 0.17.[34] The first stage is strong. Six of the seven endogenous regressors clear the Stock–Yogo 10% maximal-relative-bias value of 11.22, mostly by a wide margin: the four quintile market-income terms and the two controls return Sanderson–Windmeijer F-statistics from 18.1 to 493.5. The exception, the aggregate income-per-capita anchor ($F =$ 7.8) still clears the 20% relative-bias bar of 5.98. It is a control we never interpret. Every marginal effect we report is a strongly identified quintile term. In any case, identification does not rest on first-stage strength at all. A weak-instrument-robust Anderson–Rubin test rejects in all five disposable-income equations at $p < .001$. Finally, the difference-in-Hansen (C) test interrogates the most suspect lag. We add the two-year lag, the shortest of the deep lags, and hence the one most likely to be correlated with the current shock,

[34]This requires the heteroskedasticity-robust weight matrix. The homoskedastic weight is rejected at $p < .001$. A p close to one would itself be a warning sign of a near-singular (weak) weight matrix, not a mark of unusually valid instruments, so a comfortable pass at 0.17 is the healthier statement.

to the instrument set. The difference-in-Hansen (C) test is nowhere near rejection, so the nearest lag satisfies the exclusion restriction.

8 Directions for Further Research

It is natural to conceive of the scenarios $\{Q_i \to Q_j\}$ as independent in the sense that their outcomes are not co-determined. For instance, suppose that a transfer of market income share from the middle quintile to the top quintile is followed by an equal and opposite transfer of market income share from the top quintile to the middle quintile. In a perfectly passive environment, with all else equal, the disposable income distribution would revert to its original position, implying that the marginal effects of the initial and final market income shifts perfectly offset. In reality, however, these events (and their effects) may be completely disconnected. Indeed, if disturbances to the market income distribution can set off a fresh renegotiation of the political economy, anything (consistent with rational political coalitions and respecting economic constraints) is possible.

The existence of asymmetries would thus provide strong evidence of "active," i.e. discretionary, policy responses. Indeed, since any differentiable function mapping the market income distribution to the disposable income distribution satisfies symmetry by construction, a statistical rejection of symmetry might be interpreted as a rejection of static policy rules in general.[35] Formally, symmetry holds if, for all l, k, and j

$$\frac{\partial D^l}{\partial m(k)} = v \text{ when } m(j) \text{ is excluded} \iff \frac{\partial D^l}{\partial m(j)} = -v \text{ when } m(k) \text{ is excluded} \quad (10)$$

where $m(k)$ and D^k are quintile k's share of market and disposable income, respectively. In words, the effect on the disposable income distribution of a transfer of market income

[35] Here, we use the term 'static policy rules' to refer to fully prescriptive policy functions of (non-dynamic) state variables. In practice, policy proposals are often judged by 'rules' of a more dynamic flavor and which are better described as constraints, such as the requirement that any change be progressive (i.e. effecting lower post-tax, post-transfer inequality).

from quintile k to quintile j is equal and opposite to the effect of the reverse transfer from quintile j to quintile k.

The possibility of asymmetric effects, and hence independent experiments, path dependent outcomes, etc., gives rise to many other conjectures. For instance, we might say that monotonicity, as discussed in the empirical results Section 4, is satisfied if

1. $\frac{\partial D^l}{\partial m(l)}$ is decreasing in l (holding base fixed) and increasing in the base quintile
2. $\frac{\partial D^l}{\partial m(k)}$, $l = base$, is decreasing in l and increasing in k
3. $\frac{\partial D^l}{\partial m(k)}$, $k \neq l$ and $l \neq$ base, is decreasing in l (holding k and base fixed)

where $m(k)$ and D^k are quintile k's share of market and disposable income, respectively. In words, if market income shifts from one quintile to another, the post-tax, post-transfer gain to the receiving quintile is decreasing in the receiving quintile's rank (holding the giving quintile fixed) and increasing in the rank of the giving quintile, the post-tax, post-transfer loss to the giving quintile is increasing in the giving quintile's rank (holding the receiving quintile fixed) and decreasing in the rank of the receiving quintile, and the spillover effect on the post-tax, post-transfer income share of an unrelated quintile is decreasing in that quintile's rank. The basic idea is that the system is always less generous to higher income individuals, whatever the circumstances. This is intuitive, and in fact, we examined this hypothesis in Section 4 for a subset of the experiments, $\{Q_i \to Q_5\}_{i=1}^{4}$. However, the monotonicity property cannot be jointly satisfied by all the experiments, unless the marginal effects are also asymmetric. Quite simply, if, say, experiments$\{Q_i \to Q_5\}_{i=1}^{4}$ satisfy monotonicity, then symmetry implies that $\{Q_5 \to Q_i\}_{i=1}^{4}$ satisfy backward monotonicity (because all the signs are switched).

Thus, symmetry has many implications, and the question of whether the estimates exhibit the property is clearly of great interest. In principle, the conditions (10) form a joint hypothesis which can be statistically tested against the data, for example with a Wolak (1989) test. Unfortunately, any standard differentiable functional form regression, indeed any regression model based on a fixed mapping from the market to the disposable income distribution, implicitly subsumes symmetry by construction, making such a test infeasible. In our setting, this takes the form of cross-system restrictions (see Appendix A). The problem is mitigated somewhat by including higher order terms in the regression. In that case, the equation is nonlinear, so the choice of which base variable to drop affects the estimates. However, the differences are minor. In essence, the same information is being used to estimate the same relationships. Alternatively, we could use a unique set of instruments for each quintile/base variable (i.e. for each system of equations). As with higher order terms, this would relax symmetry in a mechanical sense. Conceptually, however, the instruments would not only have to be associated with the corresponding quintile, but "correlated" in only one direction (i.e. increases induce increases but decreases do not induce decreases, or vice versa). This is of course a tall order. Indeed, a fully unconstrained estimation requires a wholly different paradigm, e.g. a dynamic setup with possible path dependence.

In addition to these modeling and econometric issues, considerations beyond current market and disposable incomes, as discussed in Section 1, may provide valuable extensions to our work.

A different extension concerns the gap between actual and perceived inequality. Our framework, like the median-voter tradition it builds on, treats voters as responding to the actual market-income distribution. Yet people are frequently mistaken about the level and trend of inequality and about their own place in it, and cross-national evidence indicates that the demand for redistribution tracks *perceived* rather than actual inequality (Gimpelson and Treisman, 2018). The mapping from inequality to redistributive preferences appears to run through fairness judgments and salient reference groups, co-workers, neighbors, peers, rather than the aggregate distribution (Hvidberg, Kreiner, and Stantcheva, 2023), and through beliefs about mobility and how policy works more generally (Stantcheva, 2024). We estimate the response to actual inequality, and we read our state-level estimates as informative precisely because inequality is arguably more locally observable than in the cross-national settings where the actual-to-demand link is weakest. The perceptual channel nonetheless bounds any structural reading of the median-voter mechanism and is a natural direction for future work.

9 Conclusion

This paper breaks new ground in the empirics of income inequality and redistribution. Methodologically, we make two principal contributions. First, we construct richer, larger, and more comprehensive data sets at the local level over a quarter century. This enhances the accuracy and precision of the estimates, allowing us to include more control variables in the regression. What's more, the regional nature of the data makes it possible to instrument for exogenous shocks to the market income distribution (by exploiting local heterogeneity in exposure to national/international factors) to account for reverse causality.

Second, rather than use summary statistics (e.g. Gini coefficients) to measure both redistribution and inequality, we use quintile shares to represent whole distributions and a system of equations to map the market income distribution to the disposable income distribution. This approach allows us to explore a larger and more refined set of hypotheticals, distinguish between income groups, and ask questions beyond the scope of earlier studies.

By showing that taxes and transfers are most heavily responsive to the income share of the middle quintile, we provide evidence most consistent with the median voter hypothesis and Director's Law at work in the United States in our sample period. We also document a sizable real income effect, richer societies provide somewhat less bottom-quintile insurance, mainly by reshaping spillovers, and a measurable role for the history of market inequality, with higher past inequality flattening the middle-quintile peak.

We also find large, statistically significant spillover effects on quintiles whose income shares remain unchanged, a novel result which suggests that any tax and transfer system attempting to stabilize the disposable income distribution provides an incomplete model of future redistribution (either “active” policy responses play an important role, or passive feedback effects are more complicated). Finally, our examination of insurance coalitions across income quintiles would not be possible without the framework introduced in the paper.

As demonstrated in Section 7, the assumptions satisfy all relevant statistical tests, including the overidentification and difference-in-Hansen tests for the exogeneity of the instruments, as well as the weak-instrument-robust Anderson–Rubin test. What’s more, the marginal effects are quite precisely estimated, the model performs well both in and out of sample, and our main findings are robust to the choice of controls, instruments, standard errors, etc. Nevertheless, it remains to be seen whether similar patterns hold across countries, regions, time periods, political systems, as discussed in Section 8, or with even more complex regression designs.

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Appendices

A Cross-System Restrictions

Suppose the first market quintile is excluded from the regression (the lowest quintile is the "base"). Then, for the k^{th} equation,

$$\text{disp}_{it}^{k} = \varphi^{k} + \beta_2^k \,\text{mkt}(2)_{it} + \beta_3^k \,\text{mkt}(3)_{it} + \beta_4^k \,\text{mkt}(4)_{it} + \beta_5^k \,\text{mkt}(5)_{it} + \lambda^k \bar{m}_{it} + e_{it}^k, \quad (11)$$

where $\text{mkt}(i)$ is the mean market income of quintile i, $\bar{m} = \frac{1}{5}\sum_i \text{mkt}(i)$ is aggregate market income per capita, and the coefficients are chosen so that e minimizes the GMM objective (state and year effects suppressed for clarity).

Because $\bar{m}$ is the average of the five quintile means, $\text{mkt}(2) = 5\bar{m} - \text{mkt}(1) - \text{mkt}(3) - \text{mkt}(4) - \text{mkt}(5)$, so $\text{mkt}(2)$ may be eliminated in favor of $\text{mkt}(1)$:

$$\begin{aligned} \text{disp}_{it}^{k} = \varphi^{k} - \beta_2^k \,\text{mkt}(1)_{it} &+ \left[\beta_3^k - \beta_2^k\right] \text{mkt}(3)_{it} + \left[\beta_4^k - \beta_2^k\right] \text{mkt}(4)_{it} \\ &+ \left[\beta_5^k - \beta_2^k\right] \text{mkt}(5)_{it} + \left[\lambda^k + 5\beta_2^k\right] \bar{m}_{it} + e_{it}^k. \end{aligned} \quad (12)$$

Equation (12) is the same regression as (11), except that $\text{mkt}(2)$ is dropped rather than $\text{mkt}(1)$, the second quintile, rather than the first, is the base. The regressors that remain when $\text{mkt}(1)$ is excluded span the same space as those that remain when $\text{mkt}(2)$ is excluded, and there is nothing special about that choice of base. Any quintile may be eliminated in the same way. Since the same instruments and controls are used in every equation, the same residual e^k minimizes the GMM objective regardless of which quintile is excluded, so the estimates for one choice of base are completely determined by those for any other.

For example, denoting with a prime the coefficients estimated when the second quintile is excluded,

$$\beta_1^{k\prime} = -\beta_2^k$$
$$\beta_i^{k\prime} = \beta_i^k - \beta_2^k,\ i \geq 3$$
$$\lambda^{k\prime} = \lambda^k + 5\beta_2^k$$

The restriction $\beta_1^{k\prime} = -\beta_2^k$ is intuitive. The effect on the disposable income distribution of a transfer of market income from quintile 1 to quintile 2 is equal and opposite to that of a transfer from quintile 2 to quintile 1. None of these constraints is formally imposed. They arise as a by-product of the linear functional form and the freedom to re-choose the base.

B Baseline SUR Estimates

In this section, estimates are computed under the assumption that the regressors are exogenous (no instruments are used). Standard errors are robust to heteroskedasticity as well as autocorrelation.[36]

Marginal Effects

In the tables below, values represent the estimated marginal effect at mean values of a 1-percentage point transfer of market income from the base quintile to the row quintile on the disposable income share of the column quintile. P-values are calculated under the null hypothesis (7), i.e. that "own" effects are either 1 (receiving) or -1 (losing) and all other effects are zero. Symbol * indicates significance at the 10% level, ** indicates significance at the 5% level, and *** indicates significance at the 1% level.

[36] We cluster the standard errors by state, as in the main text.

Table A1: Change in Disposable Income Distribution to a Change in Market Income Distribution

Base Quintile $= Q_1$					
Experiment	ΔD_1	ΔD_2	ΔD_3	ΔD_4	ΔD_5
$M1 \rightarrow M2$	-0.455***	0.390***	-0.122*	0.093	0.095
$M1 \rightarrow M3$	-0.562***	0.289***	0.357***	-0.113	0.029
$M1 \rightarrow M4$	-0.536***	-0.031	0.048	0.734***	-0.215*
$M1 \rightarrow M5$	-0.621***	-0.027	-0.150***	-0.032	0.829***
Base Quintile $= Q_2$					
Experiment	ΔD_1	ΔD_2	ΔD_3	ΔD_4	ΔD_5
$M2 \rightarrow M1$	0.455***	-0.390***	0.122*	-0.093	-0.095
$M2 \rightarrow M3$	-0.106	-0.101	0.479***	-0.205**	-0.067
$M2 \rightarrow M4$	-0.081	-0.421***	0.171***	0.641***	-0.310***
$M2 \rightarrow M5$	-0.166***	-0.417***	-0.027	-0.124**	0.734***
Base Quintile $= Q_3$					
Experiment	ΔD_1	ΔD_2	ΔD_3	ΔD_4	ΔD_5
$M3 \rightarrow M1$	0.562***	-0.289***	-0.357***	0.113	-0.029
$M3 \rightarrow M2$	0.106	0.101	-0.479***	0.205**	0.067
$M3 \rightarrow M4$	0.025	-0.320***	-0.309***	0.847***	-0.243***
$M3 \rightarrow M5$	-0.059	-0.316***	-0.507***	0.081*	0.801***
Base Quintile $= Q_4$					
Experiment	ΔD_1	ΔD_2	ΔD_3	ΔD_4	ΔD_5
$M4 \rightarrow M1$	0.536***	0.031	-0.048	-0.734***	0.215*
$M4 \rightarrow M2$	0.081	0.421***	-0.171***	-0.641***	0.310***
$M4 \rightarrow M3$	-0.025	0.320***	0.309***	-0.847***	0.243***
$M4 \rightarrow M5$	-0.085***	0.004	-0.198***	-0.766***	1.044***
Base Quintile $= Q_5$					
Experiment	ΔD_1	ΔD_2	ΔD_3	ΔD_4	ΔD_5
$M5 \rightarrow M1$	0.621***	0.027	0.150***	0.032	-0.829***
$M5 \rightarrow M2$	0.166***	0.417***	0.027	0.124**	-0.734***
$M5 \rightarrow M3$	0.059	0.316***	0.507***	-0.081*	-0.801***
$M5 \rightarrow M4$	0.085***	-0.004	0.198***	0.766***	-1.044***

Notes: The table shows the marginal effects on disposable income shares when a base quintile loses one percent of market income share to a receiving quintile (rows). Estimates are from the SUR specification. Significance: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Insurance Coalition

Table A2: Insurance Coalition Frequency Matrix

Quintiles	Q1	Q2	Q3	Q4	Q5
Q1	NaN	0.1	0.3	0.2	0.1
Q2	0.1	NaN	0.3	0.1	0.2
Q3	0.3	0.3	NaN	0.2	0.1
Q4	0.2	0.1	0.2	NaN	0.1
Q5	0.1	0.2	0.1	0.1	NaN

Notes: The table reports the frequency with which each income quintile forms part of an insurance coalition with other quintiles. Each off-diagonal entry is the fraction of the twenty redistribution experiments in which the two quintiles' disposable-income shares move significantly in the same direction, relative to the lump-sum benchmark in (7). Diagonal entries (NaN) are undefined because a quintile cannot form a coalition with itself.

C Baseline GMM Estimates

In this section, the market income shares as well as the control variables for market income per capita, the cyclical unemployment rate, and the poverty rate are assumed endogenous, and the variables defined in Section 3.3 (Bartik cumulative shock, import exposure, the commodity interactions) as well as lags of the endogenous variables are used as instruments. Standard errors are robust to heteroskedasticity as well as autocorrelation.[37]

[37] We cluster the standard errors by state, as in the main text.

Marginal Effects

Table A3: Change in Disposable Income Distribution to a Change in Market Income Distribution

Base Quintile $= Q_1$					
Experiment	ΔD_1	ΔD_2	ΔD_3	ΔD_4	ΔD_5
$M1 \to M2$	-1.021***	0.675***	0.037	0.452**	-0.143
$M1 \to M3$	-1.229***	0.225*	0.270***	0.290**	0.444**
$M1 \to M4$	-0.785***	0.139	0.094	1.116***	-0.564**
$M1 \to M5$	-1.125***	0.107	-0.090	0.279**	0.829***
Base Quintile $= Q_2$					
Experiment	ΔD_1	ΔD_2	ΔD_3	ΔD_4	ΔD_5
$M2 \to M1$	1.021***	-0.675***	-0.037	-0.452**	0.143
$M2 \to M3$	-0.209	-0.450***	0.233**	-0.162	0.588*
$M2 \to M4$	0.236	-0.536***	0.056	0.664***	-0.420*
$M2 \to M5$	-0.105	-0.568***	-0.127*	-0.173	0.972***
Base Quintile $= Q_3$					
Experiment	ΔD_1	ΔD_2	ΔD_3	ΔD_4	ΔD_5
$M3 \to M1$	1.229***	-0.225*	-0.270***	-0.290**	-0.444**
$M3 \to M2$	0.209	0.450***	-0.233**	0.162	-0.588*
$M3 \to M4$	0.444**	-0.086	-0.177**	0.826***	-1.008***
$M3 \to M5$	0.104	-0.118*	-0.360***	-0.011	0.385**
Base Quintile $= Q_4$					
Experiment	ΔD_1	ΔD_2	ΔD_3	ΔD_4	ΔD_5
$M4 \to M1$	0.785***	-0.139	-0.094	-1.116***	0.564**
$M4 \to M2$	-0.236	0.536***	-0.056	-0.664***	0.420*
$M4 \to M3$	-0.444**	0.086	0.177**	-0.826***	1.008***
$M4 \to M5$	-0.340***	-0.031	-0.184***	-0.837***	1.393***
Base Quintile $= Q_5$					
Experiment	ΔD_1	ΔD_2	ΔD_3	ΔD_4	ΔD_5
$M5 \to M1$	1.125***	-0.107	0.090	-0.279**	-0.829***
$M5 \to M2$	0.105	0.568***	0.127*	0.173	-0.972***
$M5 \to M3$	-0.104	0.118*	0.360***	0.011	-0.385**
$M5 \to M4$	0.340***	0.031	0.184***	0.837***	-1.393***

Notes: The table shows the marginal effects on disposable income shares when quintile i loses one percentage point of market income share to quintile j ($M_i \to M_j$). Estimates are from the baseline GMM specification. Significance: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

C.1 Insurance Coalition

Table A4: Insurance Coalition Frequency Matrix

Quintiles	**Q1**	**Q2**	**Q3**	**Q4**	**Q5**
Q1	NaN	0.0	0.1	0.2	0.1
Q2	0.0	NaN	0.3	0.2	0.2
Q3	0.1	0.3	NaN	0.2	0.3
Q4	0.2	0.2	0.2	NaN	0.2
Q5	0.1	0.2	0.3	0.2	NaN

Notes: The table reports the frequency with which each income quintile forms part of an insurance coalition with other quintiles. Each off-diagonal entry is the fraction of the twenty redistribution experiments in which the two quintiles' disposable-income shares move significantly in the same direction, relative to the lump-sum benchmark in (7). Diagonal entries (NaN) are undefined because a quintile cannot form a coalition with itself.

C.2 Elasticities

In the tables below, values represent the estimated elasticities at mean values of a 1-percentage point transfer of market income from the base quintile to the row quintile on the disposable income share of the column quintile. P-values are calculated under the null hypothesis (9) described in Section 5.1. Symbol * indicates significance at the 10% level, ** indicates significance at the 5% level, and *** indicates significance at the 1% level.

Table A5: Elasticities from GMM Estimates

Base Quintile $= Q_1$					
Experiment	Q_1	Q_2	Q_3	Q_4	Q_5
$M1 \rightarrow M2$	-1.444	0.515***	0.020	0.182**	-0.032
$M1 \rightarrow M3$	-2.961	0.292*	0.252***	0.198**	0.167**
$M1 \rightarrow M4$	-2.864	0.273	0.132	1.158	-0.321**
$M1 \rightarrow M5$	-8.425	0.433	-0.260	0.593**	0.968
Base Quintile $= Q_2$					
Experiment	Q_1	Q_2	Q_3	Q_4	Q_5
$M2 \rightarrow M1$	0.423***	-0.151***	-0.006	-0.053**	0.009
$M2 \rightarrow M3$	-0.502	-0.585***	0.217***	-0.111	0.221*
$M2 \rightarrow M4$	0.860	-1.055***	0.080	0.689**	-0.239*
$M2 \rightarrow M5$	-0.784	-2.292***	-0.368*	-0.368	1.135
Base Quintile $= Q_3$					
Experiment	Q_1	Q_2	Q_3	Q_4	Q_5
$M3 \rightarrow M1$	0.509***	-0.050*	-0.043***	-0.034**	-0.029**
$M3 \rightarrow M2$	0.295	0.343***	-0.127***	0.065	-0.130*
$M3 \rightarrow M4$	1.621**	-0.170	-0.249***	0.857	-0.573***
$M3 \rightarrow M5$	0.778	-0.475*	-1.042***	-0.023	0.449***
Base Quintile $= Q_4$					
Experiment	Q_1	Q_2	Q_3	Q_4	Q_5
$M4 \rightarrow M1$	0.325***	-0.031	-0.015	-0.132***	0.036**
$M4 \rightarrow M2$	-0.334	0.409***	-0.031	-0.267***	0.093*
$M4 \rightarrow M3$	-1.070**	0.112	0.164***	-0.566*	0.379***
$M4 \rightarrow M5$	-2.548***	-0.127	-0.531***	-1.782	1.625***
Base Quintile $= Q_5$					
Experiment	Q_1	Q_2	Q_3	Q_4	Q_5
$M5 \rightarrow M1$	0.466***	-0.024	0.014	-0.033**	-0.054***
$M5 \rightarrow M2$	0.148	0.433***	0.069*	0.070	-0.215
$M5 \rightarrow M3$	-0.250	0.153*	0.335***	0.008	-0.145***
$M5 \rightarrow M4$	1.242***	0.062	0.259***	0.869	-0.792***

Notes: Each cell is the elasticity of quintile l's disposable income share with respect to quintile k's market income share, $e_k^l = (m_k/D_l)\,\partial D^l/\partial m_k$, evaluated at sample-mean shares and computed from the recovered share marginal effects in Table A3. Rows are labelled by the receiving quintile k. Columns are the affected quintile l. Significance markers test each elasticity against the mechanical fixed net-tax null in (9), using delta-method standard errors from the GMM cluster-robust covariance: ${}^{*}p < 0.10$, ${}^{**}p < 0.05$, ${}^{***}p < 0.01$ (two-sided). The pervasive rejections indicate that redistribution departs systematically from a mechanical fixed-tax schedule.

D Charts and Tables

Table A6: Market Income Share Changes by Quintile, 1991-2013 (%)

State	Q1	Q2	Q3	Q4	Q5
Alabama	-0.534	-1.267	-2.061	-1.897	5.779
Alaska	-0.430	0.105	-0.138	-0.425	0.880
Arizona	-1.532	-2.827	-2.873	-1.078	8.288
Arkansas	-0.358	-1.982	-0.875	-0.263	3.463
California	0.197	-1.255	-2.169	-2.318	5.558
Colorado	-0.122	-0.775	-1.034	-1.662	3.585
Connecticut	-1.449	-2.757	-2.016	-0.096	6.331
Georgia	-0.107	-1.060	-1.872	-1.229	4.281
Idaho	-0.767	-1.723	-0.548	0.646	2.331
Illinois	-0.101	-1.163	-0.997	-0.463	2.706
Indiana	-0.806	-0.751	-0.426	-0.468	2.445
Iowa	-1.173	-1.710	-0.809	0.142	3.481
Kansas	-1.272	-1.730	-0.578	1.423	2.116
Kentucky	0.368	-1.089	-2.494	-1.335	4.549
Louisiana	-0.055	-1.797	-1.904	-0.670	4.433
Maine	-0.647	-1.736	-1.721	0.616	3.460
Massachusetts	-1.143	-1.810	-0.674	-0.681	4.295
Michigan	0.268	-1.426	-1.778	0.353	2.581
Minnesota	0.347	-0.051	0.290	0.190	-0.776
Mississippi	-0.720	-1.680	-1.564	-1.914	5.780
Missouri	-0.650	-1.301	-1.307	-0.126	3.394

State	Q1	Q2	Q3	Q4	Q5
Montana	-0.408	-0.981	-1.059	-0.551	2.946
Nevada	-1.261	-2.289	-2.345	-0.383	6.309
New Hampshire	-1.188	-1.861	-1.251	0.214	4.035
New Jersey	-0.764	-2.221	-1.850	-0.990	5.827
New York	0.073	-1.441	-2.066	-1.271	4.708
North Carolina	-1.090	-2.258	-1.631	-0.709	5.683
North Dakota	-0.286	-1.244	-0.978	-1.015	3.409
Ohio	-0.517	-1.813	-1.413	-0.168	3.919
Oklahoma	0.146	-0.010	0.016	-1.640	1.427
Oregon	-1.095	-2.553	-2.412	-0.938	6.999
Pennsylvania	-0.888	-1.574	-1.036	0.301	3.202
Rhode Island	-0.810	-2.838	-1.798	0.242	5.213
Texas	-0.043	-0.597	-1.173	-1.203	3.001
Utah	0.165	-0.716	-0.979	-0.150	1.650
Vermont	-0.753	-1.989	-2.030	-1.049	5.852
Virginia	-0.849	-0.680	0.413	0.611	0.509
Washington	-1.083	-2.775	-2.993	-1.956	8.816
West Virginia	-0.552	-1.730	-2.442	-2.229	6.928
Wisconsin	-1.221	-1.551	-1.621	-0.744	5.089
Wyoming	-0.668	-1.465	-1.077	0.834	2.367

Notes: The table reports changes in market income shares by quintile for each U.S. state from 1991 to 2013. Each row displays the percentage change in income share for quintiles Q1-Q5. Positive values indicate gains in income share, while negative values indicate losses.

E Derivation of (9) in Section 5.1

A value of 1 in the first case follows from first degree homogeneity of a quintile's disposable income share in its own market income share. Letting D and M denote aggregate disposable and market income, respectively, and letting ϕ_i denote the net tax rate applied uniformly to quintile i, we have $P(l) = (1-\phi_l)\, X_l \frac{M}{D}$, which is homogenous in X_l (holding M and D fixed). Even if we suppose that tax rates are idiosyncratic within quintiles, $P(l) = \frac{\sum_i m_{li}(1-\phi_{li})}{D}$, where m_{li} and ϕ_{li} are the market income and net tax rate, respectively, of individual i in quintile l. This is linearly homogenous in $\overrightarrow{m_l}$, and hence linearly homogenous in X_l if $\overrightarrow{m_l}$ is linearly homogenous in X_l (i.e. proportional changes in market income are uniform within quintiles).

A value of $\frac{-P(k)}{P(l)}$ in the second case follows from the fact that if we assume that the disposable income shares of quintiles not involved in the transfer are unaffected (i.e.

$e_k^l = 0$ for $l \notin \{k, base\}$, as in the null hypothesis above), then the sum of the disposable income shares of the receiving and losing quintiles is fixed, i.e. $P(k) + P(base) = C$, C a constant. Thus,

$$\frac{X_k}{P(k)\,P(base)}\left[\frac{\partial P(k)}{\partial X_k} + \frac{\partial P(base)}{\partial X_k}\right] = 0$$

$$\frac{1}{P(base)} + \frac{e_k^{base}}{P(k)} = 0$$

$$e_k^{base} = -\frac{P(k)}{P(base)}$$

where the second line uses $e_k^k = 1$, established in the previous paragraph.

F Income Definitions

Table A7: Comparison of Income Definitions Across Studies

General Setup	CBO	Census P-60	Census Research Series (2005)	Our Work	Piketty–Saez	Armour et al.
Data Sources	CPS, SOI (full), CES	CPS	CPS, SOI, AHS, SIPP	CPS + Census imputations	IRS tables	CPS + Census imputations, SCF, HPI
Income Unit	Household	Household	Household	Household/Family	Tax Unit	Household
Equivalence Scale	Square root	Betson	Betson	Betson/Eurostat	None	Square root
Unit of Analysis	Individual	Household	Household	Individual	Tax Unit	Individual
Price Level Adjustment (Temporal)	PCE deflator	CPI-U-RS	CPI-U-RS	None (nominal)	CPI-U	CPI-U-RS
Market Income						
Labor earnings: wages and salaries, self-employment	Y	Y	Y	Y	Y	Y
Employer-paid health insurance	Y	Y	Y			
Employer's share of payroll taxes	Y					
Corporate tax borne by workers	Y					
Work expenses (deduction)	Y					
Capital Income (Privately Held)						
Business income	Y	Y	Y	Y	Y	Y
Interest	Y	Y	Y	Y	Y	Y
Dividends	Y	Y	Y	Y	Y	Y
Rental income	Y	Y	Y	Y	Y	Y
Imputed rent for owner-occupied housing	Y	Y				
Corporate tax borne by capital owners	Y					
Retirement and Related Income						
Pensions	Y	Y	Y	Y	Y	Y
401(k) and IRA withdrawals	Y	Y	Y	Y	Y	Y
Survivors' income	Y	Y	Y	Y	Y	Y
Disability income	Y	Y	Y	Y	Y	
Capital Gains						
Taxable realized capital gains	Y	Y	Y			
Accrued capital gains	Y					
Private Transfers						
Remittances	Y					
Alimony	Y	Y	Y	Y	Y	
Child support	Y	Y	Y	Y		
Scholarships, grants, etc.	Y	Y	Y	Y		
Transfers – Cash						
Social Security	Y	Y	Y	Y	Y	
Unemployment insurance	Y	Y	Y	Y	Y	Y
Supplemental Security Income (SSI)	Y	Y	Y	Y	Y	
Welfare (means-tested cash assistance)	Y	Y	Y	Y	Y	
Veterans' benefits	Y	Y	Y	Y	Y	
Workers' compensation	Y	Y	Y	Y	Y	
Educational assistance (cash)	Y	Y	Y	Y		
Transfers – In-kind						
SNAP (food stamps)	Y	Y	Y	Y		
School meals	Y	Y	Y	Y		
Housing assistance	Y	Y	Y	Y		
Energy assistance	Y	Y				
Medicare	Cost		Cost	Cost		
Medicaid	Cost		Cost	Cost		
CHIP	Cost					
Taxes – Federal						
Income tax before credits	Y	Y	Y	Y		
Earned Income Tax Credit (EITC)	Y	Y	Y	Y		
Other tax credits	Y	Y	Y			
Employee's share of payroll taxes	Y	Y	Y	Y		
Employer's share of payroll taxes	Y	Y	Y	Y	Y	Y
Corporate tax borne by workers	Y	Y	Y	Y	Y	Y
Corporate tax borne by capital	Y	Y	Y	Y	Y	Y
Excise taxes	Y					
Taxes – State and Local						
State taxes, before credits	Y					
State taxes, after credits	Y	Y				
Local property taxes	Y	Y				

Notes: This table compares the construction of income concepts across several prominent data sources and studies. A checkmark (Y) indicates that the corresponding income component or tax item is included in the definition used by the given source. When labels such as "Cost" or "Fungible value" appear for Medicare, Medicaid, and CHIP, they indicate the valuation method used for in-kind health benefits. Because this LaTeX version is reconstructed from an image, you may wish to double-check the placement of checkmarks against the original table.